Oscillation Suppression-Enhanced Cooperative Control via Refined Cooperative Disturbance Estimation for Aerial Co-Transportation System

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***Abstract*— This article focuses on the oscillation suppression-enhanced cooperative control design for the aerial co-transportation system consisting of two quadrotors and a tethered pipe. The system dynamics are analyzed in depth, which yields a decoupled model under multiple disturbances by utilizing the variation linearization technique and coordinate transformations. Based on this model, a refined cooperative disturbance estimation strategy is developed to capture the angle dynamics of the cables without direct measurements of swing angles. Then the estimation results are used for designing a cooperative control law to guarantee the performance in rapid suppression of the payload oscillation and in accurate positioning of the quadrotors under system uncertainties. The stability and convergence of the overall system is established using Lyapunov theory. Finally, experiments validate and demonstrate the superiority of the proposed method over the existing ones.**

***Note to Practitioners*—This paper is motivated by the requirement of safe control schemes for aerial co-transportation systems. The unexpected oscillation of the payload may result in serious accidents, and therefore efficiently suppressing the payload swing is the main concern of the research. Nevertheless, the cascaded underactuation property and the complicated couplings among the drones make it difficult to directly control the payload. Up till now, at the cost of additional weight and more complicated structure, most existing methods relying on extra sensors to detect the states of the payload for feedback control. Accounting for the foregoing problems, this article presents a novel sensorless control scheme for suppressing the payload oscillation. The cable angles are estimated using only the states of the drones. Moreover, cooperative control laws are designed based on the estimated results so that both antiswing and positioning performance are guaranteed. All these aspects are verified by rigorous theoretical analysis and hardware experiments.**

***Index Terms*— Cooperative transportation, oscillation suppression, cooperative control, refined cooperative disturbance estimation.**

Received 12 June 2024; revised 16 August 2024; accepted 19 October 2024. Date of publication 31 October 2024; date of current version 26 March 2025. This article was recommended for publication by Associate Editor C. Zhang and Editor M. Dotoli upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 62173024, Grant 62173028, Grant 62233015, and Grant 62273024; in part by Zhejiang Natural Science Foundation under Grant LZ22F030012; in part by the Innovation Team of University in Guizhou Province under Grant 2022033; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 024A1515011493; in part by the Program for Changjiang Scholars and Innovative Research Team under Grant IRT_16R03; in part by the Science, Technology and Innovation Project of Xiongan New Area under Grant 2023XAGG0062; in part by the Near Space Technology and Industry Guidance Fund under Grant LKJJ-2023008-01; and in part by Beijing Natural Science Foundation under Grant 4232060 and Grant IS23065. *(Corresponding author: Hao Lu.)*

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Digital Object Identifier 10.1109/TASE.2024.3485237

I. Introduction

DRONE delivery has achieved significant development in the past few years because it offers a faster and more versatile solution compared with ground transportation. To increase the load capacity of the aerial transportation system, one can either use a large aerial vehicle or employ more drones to transport a payload cooperatively. Due to the limited capability of a single aerial vehicle, cooperative transportation by more vehicles is proved to be a more preferable choice, which has some apparent advantages, such as higher payload capacity [1], lower overall cost [2], and inherent mission redundancy [3]. The aerial co-transportation systems have gained extensive adoption across a variety of fields. In the research [4], three quadrotors were tasked with transporting a point-mass payload over a 45-meter outdoor span, with the assistance of air dampers to bolster the system stability. Additionally, a fleet of four Crazyflie drones was employed to collaboratively carry a white fabric in a decentralized manner, as detailed in the study [5]. Moreover, aerial co-transportation systems have carved out a specialized role in the realm of precision agriculture. As highlighted in the work [6], a pair of quadrotors, equipped with a fluid tank and a pump, were utilized to haul a pipe with nozzles spraying pesticide on crops.

The number of drones used in aerial transportation depends on the payload scale and the task objective, e.g., manipulating the payload attitude or simply transporting the payload from one place to another. For the slung-load transportation, under the condition that the payload mass is within the load capacity range of the drones, using two drones to transport a one-dimensional rigid body, such as a rod or a

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beam, can be cost-effective and energy-efficient. Since the one-dimensional rigid body does not have a rotation degree of freedom around the body axis, and therefore two drones are enough to achieve pose tracking of a rod-like object [7]. Generally, for a two-dimensional or three-dimensional rigid body, at least three drones are needed to achieve the full-pose control [8]. Considering that transporting the rod-like object is a common requirement in practice, we investigate the scenario of two quadrotors cooperatively transporting a slender pipe.

The use of more drones complicates the dynamic structure of the overall system than the single-drone case and exacerbates the problems inherent in aerial transportation such as payload oscillation [9], communication delay [10], suspension failures [11], and obstacle collision [12]. Among these problems, payload oscillation is common and serious in aerial cable-suspended systems [13]. The cascaded underactuation characteristics of the suspended transportation system makes it extremely difficult to suppress the payload swing, especially with two-drone case [13]. Although using three or more drones can relieve the payload swing to some extent, it is of low efficiency to transport a rod due to loss of lift force. Compared with the payload swing problem in the one-drone case [14], two-drone collaborative transportation brings about the additional coupling and interference between the drones. By exploiting the coupling and interference among the two drones, this paper aims to better attenuate the payload swing for aerial co-transportation systems.

For the aerial transportation system, the existing strategies proposed to reduce the payload oscillation can be broadly classified into two categories: open-loop and closed-loop strategies. Input shaping [15], [16], [17] and trajectory planning [18], [19], [20] are verified to be two types of efficient open-loop methods in the experiments. For the single-drone-payload case in [15] and [16], input shapers are utilized to generate payload swing-free reference command, while for the two-drone-payload case in [17], input shapers are introduced to reduce both the residual oscillations of the payload and the follower. As far as trajectory planning is concerned, dynamic programming [18] and reinforcement learning [19] technique are employed to optimize the trajectory to generate payload swing-free maneuver for single-drone-payload system. In addition, a trajectory planning method based on a value-function approximation algorithm is proposed in [20] to address the problem of cable-suspended load transportation using three quadrotors such that the payload can be transported smoothly from the starting location to the target location. Although these feedforward methods can effectively reduce the swing of the payload caused by the movement of the drones, oscillations excited by the initial cable angle offset, wind disturbance, or the sudden collisions due to interaction of the payload with the environment can not be dealt with appropriately, which may lead to the instability of the system.

In terms of closed-loop methods, the authors in [21] design a nonlinear hierarchical control (NHC) law based on the energy storage function for a single drone to asymptotically stabilize the quadrotor-payload system. Similar to this method, an interconnection and damping assignment-passivity based control (IDA-PBC) strategy is successfully applied in [22] for a single drone to attenuate the payload's swing. The essence of these methods is damping injection and energy dissipation. Following this core idea, the authors in [23] develop a time-domain passivity observer (PO) for the three-drone-payload system to detect the energy flow and use an adaptive damping term to ensure that the storage function of the whole system is continuously decreasing, which guarantees the bounded stability in the experiment. In addition, a tension estimation-based position control strategy is proposed in [24] to achieve formation tracking (FT) for three-drone-payload case in the experiment.

To suppress the payload oscillation in a more refined and efficient way, several methods have introduced the cable angle information or the payload position in feedback control [16], [25], [26], [27]. For single-drone-payload case, the deflection angle of the cable attached to the load is measured by a magnetic encoder in [16] to implement the delayed feedback approach, which can effectively deal with the perturbations on the payload. For two-drone-payload case, an energy-based control strategy is employed in [25] to suppress the cable swing, where the motion capture system is utilized to obtain the payload position. In addition, GPS modules [26] and onboard cameras [27] are equipped on the aerial transportation systems for implementing the so-called load priority design, which can directly manipulate the position of the payload.

Instead of installing extra sensors for cable-related or payload-related measurement, an almost globally convergent nonlinear observer is designed in [28] to estimate the full state of the tethered drone for trajectory tracking control, only using conventional onboard sensors as an accelerometer and a gyroscope. Similarly, a state observer is established in [29] to estimate the swing angle of the cable for precise trajectory tracking of the drone. In addition, a filtering technique based on a set of recursive equations is proposed in [30] to estimate the swing angle without relying on extra sensors. However, the aforementioned methods [28], [29], [30] can only be applied to the single-drone-payload case since they do not consider the coupling and interference between the two drones caused by the suspended payload. Furthermore, only simulations are made in these articles, and model uncertainties and disturbances in engineering practice are not taken into consideration. For the single-drone-payload case in [31], the cable force on the drone is treated as a type of periodic disturbance, and is estimated through a known-frequency disturbance observer. This disturbance estimate is used directly to compensate the disturbance via the so-called multiple observers based anti-disturbance control (MOBADC), the effectiveness of which is validated by a flight test of trajectory tracking, but swing attenuation of the payload is not considered. To the best of our knowledge, there is no existing method which can estimate the swing angles of the cables in two-drone cooperative transportation under model uncertainties without direct measurement of the payload.

In this article, an aerial co-transportation system consisting of two quadrotors and a tethered pipe is considered, and a refined cooperative disturbance estimation strategy is proposed to estimate the directions of the cables only using the motion measurement of the quadrotors. Based on the estimates, the

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cooperative controllers are designed to stabilize the whole aerial co-transportation system. The main contributions are summarized in the following aspects:

1) Instead of treating the cable force on the quadrotor as the periodic disturbance as in [31], the proposed strategy establishes the detailed cable force dynamics on the two quadrotors, corresponding to "refined". Then, the variation-based linearization technique and coordinate transformations are conducted to transform the coupled system dynamics into a decoupled model for estimation and control purpose.
2) Based on the decoupled linear model, a new refined cooperative disturbance estimation strategy is developed for the first time to estimate the cable directions despite of multiple disturbances. The local swing observer uses the exchanged information of the estimate and the states from the neighbor quadrotor, corresponding to "cooperative".
3) Utilizing the estimated cable-direction information, a cooperative control law is designed to simultaneously achieve accurate quadrotor positioning and efficient payload swing suppression. To our best knowledge, such swing-angle estimation-based control law for cooperative transportation by two drones has not been reported in previous studies. Real-world flight tests are carried out to verify the effectiveness of the proposed algorithm. The test results show that the proposed method achieves better performance than existing methods in payload swing suppression.

II. Modeling and Simplification

A. Mathematical Preliminaries

For attitude representation of a rigid body, rotation matrix $\boldsymbol{\mathcal{A}} \in \mathsf{SO}(3)$ is utilized, where

$$\mathsf{SO}(3) = \left\{\boldsymbol{\mathcal{A}} \in \mathbb{R}^{3\times 3} | \boldsymbol{\mathcal{A}}^\top \boldsymbol{\mathcal{A}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}^\top = \boldsymbol{I}_3, \det(\boldsymbol{\mathcal{A}}) = 1\right\}$$

denotes the special orthogonal group [32]. Here $\boldsymbol{I}_3 \in \mathbb{R}^{3\times 3}$ is the identity matrix and $\det(\boldsymbol{\mathcal{A}})$ denotes the determinant of the matrix $\boldsymbol{\mathcal{A}}$.

The set of 3×3 skew-symmetric matrix is denoted as

$$\mathfrak{so}(3) = \{\boldsymbol{\mathcal{B}} \in \mathbb{R}^{3\times 3} | \boldsymbol{\mathcal{B}}^\top = -\boldsymbol{\mathcal{B}}\},$$

which corresponds to the tangent space of $\mathsf{SO}(3)$ at the identity $\boldsymbol{I}_3 \in \mathsf{SO}(3)$, i.e., the Lie algebra of $\mathsf{SO}(3)$.

To describe the direction of a cable, unit vector $\boldsymbol{q} \in \mathsf{S}^2$ is adopted, where

$$\mathsf{S}^2 = \{\boldsymbol{q} \in \mathbb{R}^3 | \boldsymbol{q}^\top \boldsymbol{q} = 1\}$$

denotes the unit sphere manifold embedded in $\mathbb{R}^3$ [32].

The angular velocity of a cable associated with the direction $\boldsymbol{q} \in \mathsf{S}^2$ lies in the tangent space $\mathsf{T}_{\boldsymbol{q}}\mathsf{S}^2$, where

$$\mathsf{T}_{\boldsymbol{q}}\mathsf{S}^2 = \{\boldsymbol{h} \in \mathbb{R}^3 | \boldsymbol{h}^\top \boldsymbol{q} = 0\}.$$

The Euclidean norm of a matrix $\boldsymbol{\mathcal{C}} \in \mathbb{R}^{m\times n}$ is defined as

$$\|\boldsymbol{\mathcal{C}}\| = \sqrt{\lambda_{\max}(\boldsymbol{\mathcal{C}}^\top \boldsymbol{\mathcal{C}})},$$

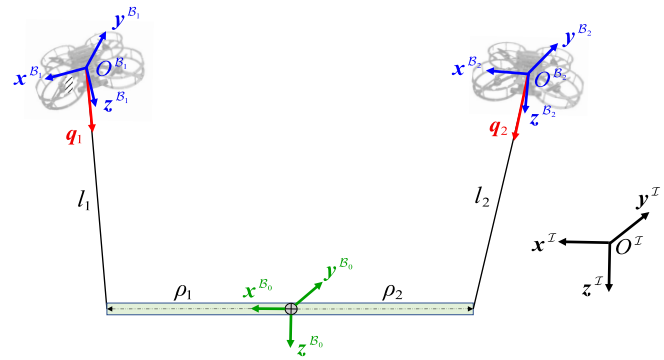

Fig. 1. Schematic of the collaborative transportation system.

where $\lambda_{\max}(\cdot)$ denotes the largest eigenvalue of the matrix.

For vector $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^\top$, we define the mapping $(\cdot)^\times : \mathbb{R}^3 \to \mathfrak{so}(3)$ as

$$\boldsymbol{\omega}^\times = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix},$$

and its inverse mapping $(\cdot)^\vee : \mathfrak{so}(3) \to \mathbb{R}^3$ as

$$(\boldsymbol{\omega}^\times)^\vee = \begin{bmatrix} \omega_x & \omega_y & \omega_z \end{bmatrix}^\top.$$

By utilizing the differentiable manifolds such as $\mathsf{SO}(3)$ and S^2 for configuration description, the models considered in this paper are expressed in a coordinate-free fashion to avoid singularities and complexities that are associated with local parameterizations [33], e.g., Eq. (3) and Eq. (4).

B. Configuration Description

We consider two quadrotors carrying a pipe through two massless cables as shown in Fig. 1, where the cables are assumed to be attached to the center of mass (CoM) of the quadrotors. Throughout this paper, the variables related to the payload are denoted by the subscript 0, and the variables for the ith quadrotor are denoted by the subscript $i \in \{1, 2\}$. The unit vectors are defined as $\boldsymbol{e}_1 = [1, 0, 0]^\top$, $\boldsymbol{e}_2 = [0, 1, 0]^\top$, and $\boldsymbol{e}_3 = [0, 0, 1]^\top$. We choose the north-east-down (NED) frame as the inertial frame $\mathcal{F}^{\mathcal{I}} = \{O^{\mathcal{I}}, \boldsymbol{x}^{\mathcal{I}}, \boldsymbol{y}^{\mathcal{I}}, \boldsymbol{z}^{\mathcal{I}}\}$, with $O^{\mathcal{I}}$ fixed at a point on the ground. The body-attached frame $\mathcal{F}^{\mathcal{B}_0} = \{O^{\mathcal{B}_0}, \boldsymbol{x}^{\mathcal{B}_0}, \boldsymbol{y}^{\mathcal{B}_0}, \boldsymbol{z}^{\mathcal{B}_0}\}$ for the payload is defined, with its origin $O^{\mathcal{B}_0}$ at the CoM of the payload; $\boldsymbol{x}^{\mathcal{B}_0}$ axis points along the pipe toward the attachment point of the first cable; $\boldsymbol{z}^{\mathcal{B}_0}$ axis is perpendicular to $\boldsymbol{x}^{\mathcal{B}_0}$, locates in the vertical plane, and points downward; $\boldsymbol{y}^{\mathcal{B}_0}$ axis completes the right-hand frame. The body-attached frame for the ith quadrotor is denoted by $\mathcal{F}^{\mathcal{B}_i} = \{O^{\mathcal{B}_i}, \boldsymbol{x}^{\mathcal{B}_i}, \boldsymbol{y}^{\mathcal{B}_i}, \boldsymbol{z}^{\mathcal{B}_i}\}$ as shown in Fig. 1. The cables are assumed taut and the lengths of the cables are assumed fixed. For the sake of the following discussion, some symbols presented in this article are summarized in Table I.

Since the cables are taut and of fixed lengths, the relationship between the CoM position of the payload $\boldsymbol{p}_0$ and that of the ith drone $\boldsymbol{p}_i$ is

$$\begin{aligned} \boldsymbol{p}_i &= \boldsymbol{p}_0 + (-1)^{i+1}\rho_i \boldsymbol{R}_0 \boldsymbol{e}_1 - l_i \boldsymbol{q}_i \\ &= \boldsymbol{p}_0 + \boldsymbol{R}_0 \boldsymbol{\rho}_i - l_i \boldsymbol{q}_i, \quad i = 1, 2, \end{aligned} \tag{1}$$

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TABLE I
NOMENCLATURE

Symbols	Physical interpretations
$\boldsymbol{d}_{\boldsymbol{p}_0}, \boldsymbol{d}_{\boldsymbol{p}_i} \in \mathbb{R}^3$	disturbance force on the payload, ith quadrotor
$\mathcal{F}^{\mathcal{I}}, \mathcal{F}^{\mathcal{B}_0}, \mathcal{F}^{\mathcal{B}_i}$	inertial frame, payload frame, ith quadrotor frame
$\boldsymbol{d}_{\boldsymbol{R}_0}, \boldsymbol{d}_{\boldsymbol{R}_i} \in \mathbb{R}^3$	disturbance moment on the payload, ith quadrotor
$f_i \in \mathbb{R}^+$	thrust magnitude of the ith quadrotor
$g \in \mathbb{R}^+$	gravity constant
$\boldsymbol{J}_0, \boldsymbol{J}_i \in \mathbb{R}^{3\times 3}$	inertia matrix of the payload, ith quadrotor
$l_0 \in \mathbb{R}^+$	half length of the pipe
$l_i \in \mathbb{R}^+$	length of the ith cable
$m_0, m_i \in \mathbb{R}^+$	mass of the payload, ith quadrotor
$\boldsymbol{p}_0, \boldsymbol{p}_i \in \mathbb{R}^3$	CoM position of the payload, ith quadrotor
$\boldsymbol{q}_i \in \mathsf{S}^2$	direction of the ith cable
$\boldsymbol{R}_0, \boldsymbol{R}_i \in \mathsf{SO}(3)$	rotation matrix of the payload, ith quadrotor
$\boldsymbol{u}_i \in \mathbb{R}^3$	control force for the ith quadrotor
$\boldsymbol{\tau}_i \in \mathbb{R}^3$	torque input for the ith quadrotor
$\boldsymbol{\omega}_i \in \mathsf{T}_{\boldsymbol{q}}\mathsf{S}^2$	angular velocity of the ith cable
$\boldsymbol{\Omega}_0, \boldsymbol{\Omega}_i \in \mathbb{R}^3$	body angular rate of the payload, ith quadrotor
$\rho_i \in \mathbb{R}$	distance from the CoM of the payload to the ith cable attachment point on the payload
$\boldsymbol{\rho}_i \in \mathbb{R}^3$	vector pointing from the CoM of the payload to the ith cable attachment point on the payload

where the vector

$$\boldsymbol{\rho_i} = (-1)^{i+1}\rho_i \boldsymbol{e}_1 \in \mathbb{R}^3 \quad (i = 1, 2) \tag{2}$$

is used for notation simplicity.

In summary, the configuration of the presented system is described by the position $\boldsymbol{p}_0$ and the attitude $\boldsymbol{R}_0$ of the payload, the direction $\boldsymbol{q}_i$ $(i = 1, 2)$ of the cables pointing from the CoM of the ith quadrotor to the load, and the attitude $\boldsymbol{R}_i$ $(i = 1, 2)$ of the quadrotors, and it can be shown that this configuration evolves on the Cartesian product of the manifolds $\mathbb{R}^3 \times \mathsf{SO}(3) \times (\mathsf{S}^2 \times \mathsf{SO}(3))^2$.

C. Complete Nonlinear Model

The ith quadrotor can generate a thrust $-f_i \boldsymbol{R}_i \boldsymbol{e}_3$ $(i = 1, 2)$ with respect to the inertial frame $\mathcal{F}^{\mathcal{I}}$ and a torque $\boldsymbol{\tau}_i \in \mathbb{R}^3$ $(i = 1, 2)$ with respect to its body-fixed frame $\mathcal{F}^{\mathcal{B}_i}$.

Based on the assumption that the cables are attached to CoMs of the quadrotors, the body angular velocity of the ith quadrotor $\boldsymbol{\Omega}_i \in \mathbb{R}^3$ $(i = 1, 2)$ is only subject to the corresponding control torque $\boldsymbol{\tau}_i$ $(i = 1, 2)$ and the associated disturbance $\boldsymbol{d}_{\boldsymbol{R}_i}$ $(i = 1, 2)$. See Eq. (5) on the next page. Furthermore, the body angular velocity $\boldsymbol{\Omega}_i$ $(i = 1, 2)$ does not affect the dynamics of the payload and the cables directly. Therefore, we consider a simplified dynamic model where the attitude kinematics and dynamics of each quadrotor are ignored, i.e., the configuration of the whole system lies in the manifold $\mathbb{R}^3 \times \mathsf{SO}(3) \times \mathsf{S}^2 \times \mathsf{S}^2$. The kinematics of the multi-body system are represented as

$$\begin{aligned} \dot{\boldsymbol{p}}_0 &= \dot{\boldsymbol{p}}_0, \\ \dot{\boldsymbol{R}}_0 &= \boldsymbol{R}_0 \boldsymbol{\Omega}_0^{\times}, \\ \dot{\boldsymbol{q}}_i &= \boldsymbol{\omega}_i \times \boldsymbol{q}_i, \quad i = 1, 2, \end{aligned} \tag{3}$$

where the body angular rate of the payload is denoted by $\boldsymbol{\Omega}_0 \in \mathbb{R}^3$, and the angular velocity of the ith cable is denoted by $\boldsymbol{\omega}_i = \boldsymbol{q}_i \times \dot{\boldsymbol{q}}_i \in \mathsf{T}_{\boldsymbol{q}}\mathsf{S}^2$, satisfying $\boldsymbol{q}_i^{\top}\boldsymbol{\omega}_i = 0$ $(i = 1, 2)$.

The dynamic equations of the system are modeled as [33]

$$\begin{aligned} & m_T \ddot{\boldsymbol{p}}_0 + \sum_{i=1}^{2} m_i(-\boldsymbol{R}_0 \boldsymbol{\rho}_i^{\times}\dot{\boldsymbol{\Omega}}_0 + l_i \boldsymbol{q}_i^{\times}\dot{\boldsymbol{\omega}}_i) + \sum_{i=1}^{2} m_i \boldsymbol{R}_0 (\boldsymbol{\Omega}_0^{\times})^2 \boldsymbol{\rho}_i \\ & + m_i l_i \|\boldsymbol{\omega}_i\|^2 \boldsymbol{q}_i = m_T g \boldsymbol{e}_3 + \sum_{i=1}^{2}(\boldsymbol{u}_i + \boldsymbol{d}_{\boldsymbol{p}_i}) + \boldsymbol{d}_{\boldsymbol{p}_0}, \\ & \bar{\boldsymbol{J}}_0 \dot{\boldsymbol{\Omega}}_0 + \sum_{i=1}^{2} m_i \boldsymbol{\rho}_i^{\times} \boldsymbol{R}_0^{\top}(\ddot{\boldsymbol{p}}_0 + l_i \boldsymbol{q}_i^{\times}\dot{\boldsymbol{\omega}}_i + l_i \|\boldsymbol{\omega}_i\|^2 \boldsymbol{q}_i) \\ & + \boldsymbol{\Omega}_0^{\times} \bar{\boldsymbol{J}}_0 \boldsymbol{\Omega}_0 = \sum_{i=1}^{2} \boldsymbol{\rho}_i^{\times} \boldsymbol{R}_0^{\top}(\boldsymbol{u}_i + \boldsymbol{d}_{\boldsymbol{p}_i} + m_i g \boldsymbol{e}_3) + \boldsymbol{d}_{\boldsymbol{R}_0}, \\ & m_i l_i \dot{\boldsymbol{\omega}}_i - m_i \boldsymbol{q}_i^{\times}[\ddot{\boldsymbol{p}}_0 - \boldsymbol{R}_0 \boldsymbol{\rho}_i^{\times}\dot{\boldsymbol{\Omega}}_0 + \boldsymbol{R}_0 (\boldsymbol{\Omega}_0^{\times})^2 \boldsymbol{\rho}_i] \\ & \qquad = -\boldsymbol{q}_i^{\times}(m_i g \boldsymbol{e}_3 + \boldsymbol{u}_i + \boldsymbol{d}_{\boldsymbol{p}_i}), \quad i = 1, 2, \end{aligned} \tag{4}$$

where $m_T = \sum_{i=0}^{2} m_i$ and $\bar{\boldsymbol{J}}_0 = \boldsymbol{J}_0 - \sum_{i=1}^{2} m_i (\boldsymbol{\rho}_i^{\times})^2$ are used for notation simplicity, and $\boldsymbol{\rho}_i$ $(i = 1, 2)$ is as in (2). The control force $\boldsymbol{u}_i \triangleq -f_i \boldsymbol{R}_i \boldsymbol{e}_3 \in \mathbb{R}^3$ $(i = 1, 2)$ is regarded as the control inputs for this model (4).

Remark 1: In this model, the bandwidth of the control inputs $\boldsymbol{u}_i$ $(i = 1, 2)$ depends on the attitude loop of the quadrotors given as:

$$\begin{aligned} \dot{\boldsymbol{R}}_i &= \boldsymbol{R}_i \boldsymbol{\Omega}_i^{\times}, \quad i = 1, 2 \\ \boldsymbol{J}_i \dot{\boldsymbol{\Omega}}_i + \boldsymbol{\Omega}_i^{\times} \boldsymbol{J}_i \boldsymbol{\Omega}_i &= \boldsymbol{\tau}_i + \boldsymbol{d}_{\boldsymbol{R}_i}, \quad i = 1, 2, \end{aligned} \tag{5}$$

which are ignored in the simplified model (3) and (4). The attitude dynamics of the quadrotors are assumed to be sufficiently fast and accurate to track the desired attitude command [34]. Therefore, one can consider the control design of $\boldsymbol{u}_i$ $(i = 1, 2)$ and $\boldsymbol{\tau}_i$ $(i = 1, 2)$ separately. In this paper, we mainly focus on the design of $\boldsymbol{u}_i$ $(i = 1, 2)$ and adopt the method in [35] for tracking of the control forces.

In summary, the simplified system kinematics (3) and dynamics (4) evolving on the tangent bundle of the configuration manifold are derived, and they are described by the general coordinates $(\boldsymbol{p}_0, \boldsymbol{R}_0, \boldsymbol{q}_1, \boldsymbol{q}_2)$ and the general velocities $(\dot{\boldsymbol{p}}_0, \boldsymbol{\Omega}_0, \boldsymbol{\omega}_1, \boldsymbol{\omega}_2)$.

Remark 2: What is not considered in the model (4) is the damping force or the so-called Rayleigh dissipation force [36], which is difficult to model precisely in practice. Therefore, readers can consider that this damping force is included in the disturbance $\boldsymbol{d}_{\boldsymbol{p}_i}$ $(i = 0, 1, 2)$ or $\boldsymbol{d}_{\boldsymbol{R}_0}$.

III. Model Decoupling and Transformation

It is nontrivial to reveal the relation among the system states and how the states evolve under the effects of the control inputs $\boldsymbol{u}_i$ $(i = 1, 2)$ and the disturbances $\boldsymbol{d}_{\boldsymbol{p}_i}, \boldsymbol{d}_{\boldsymbol{R}_i}$ $(i = 1, 2)$

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based on the nonlinear coupled dynamics (4). Despite of being local approximation, linearization of the model provides us an insight into the structure of this nonlinear system and gives a simplified framework for control and estimation design. In this section, we present a method to decouple the system dynamics by linearizing the differential equations in (3) and (4) about a given desired equilibrium[1] $(\boldsymbol{p}_{0d}, \dot{\boldsymbol{p}}_{0d}, \boldsymbol{R}_{0d}, \boldsymbol{\Omega}_{0d}, \boldsymbol{q}_{id}, \boldsymbol{\omega}_{id})$ $(i = 1, 2)$, simplifying the model, and transforming the coordinates. For simplicity, we consider the ideal situation where the desired equilibrium is given by

$$\left(\boldsymbol{p}_{0d}, \boldsymbol{0}_{3\times 1}, \boldsymbol{I}_3, \boldsymbol{0}_{3\times 1}, (\boldsymbol{e}_3, \boldsymbol{0}_{3\times 1})^2\right). \tag{6}$$

At this desired equilibrium, the control input for the ith quadrotor is

$$\boldsymbol{u}_{id} = -(m_i + \frac{m_0\rho_j}{\rho_1+\rho_2})g\boldsymbol{e}_3, \quad i, j = 1, 2, \quad i \neq j, \tag{7}$$

and the disturbances $(\boldsymbol{d}_{\boldsymbol{p}_0}, \boldsymbol{d}_{\boldsymbol{R}_0}, \boldsymbol{d}_{\boldsymbol{p}_1}, \boldsymbol{d}_{\boldsymbol{p}_2})$ are all assumed zero.

Remark 3: The equilibrium of the system is not limited to the simplest configuration $\boldsymbol{R}_{0d} = \boldsymbol{I}_3$ as considered in this paper. As a matter of fact, the proposed method can be applied to any stable configuration of the aerial cable-suspended transportation system. Here the stability of the configuration can be verified by the positive definiteness of the potential matrix as shown in [1].

A. Linearized Model

In this subsection, we exploit the variation linearization techniques described in [37] to obtain the linear dynamics. The infinitesimal variations of the general coordinates are denoted by $(\delta\boldsymbol{p}_0, \delta\boldsymbol{R}_0, \delta\boldsymbol{q}_1, \delta\boldsymbol{q}_2)$ and the infinitesimal variations of the general velocities by $(\delta\dot{\boldsymbol{p}}_0, \delta\boldsymbol{\Omega}_0, \delta\boldsymbol{\omega}_1, \delta\boldsymbol{\omega}_2)$. The individual elements of these variations are represented as

$$\begin{aligned}
&\delta\boldsymbol{p}_0 = \boldsymbol{p}_0 - \boldsymbol{p}_{0d},\\
&\delta\dot{\boldsymbol{p}}_0 = \dot{\boldsymbol{p}}_0 - \dot{\boldsymbol{p}}_{0d} = \dot{\boldsymbol{p}}_0,\\
&\delta\boldsymbol{R}_0 \approx \boldsymbol{R}_{0d}\boldsymbol{\eta}_0^{\times} = \boldsymbol{\eta}_0^{\times},\\
&\delta\boldsymbol{\Omega}_0 \approx (\boldsymbol{R}_{0d}^{\top}\delta\dot{\boldsymbol{R}}_0)^{\vee} = \dot{\boldsymbol{\eta}}_0,\\
&\delta\boldsymbol{q}_i \approx \boldsymbol{\xi}_i \times \boldsymbol{q}_{id} = -\boldsymbol{e}_3^{\times}\boldsymbol{\xi}_i, \quad i = 1, 2,\\
&\delta\boldsymbol{\omega}_i \approx \boldsymbol{q}_{id} \times \delta\dot{\boldsymbol{q}}_i = -\boldsymbol{e}_3^{\times}\boldsymbol{e}_3^{\times}\dot{\boldsymbol{\xi}}_i, \quad i = 1, 2,
\end{aligned} \tag{8}$$

where the vectors $\boldsymbol{\eta}_0 \in \mathbb{R}^3$ and $\boldsymbol{\xi}_i \in \mathbb{R}^3$ $(i = 1, 2)$ are treated as linear approximations of the errors between the desired and actual states of the system on $\mathsf{SO}(3)$ and S^2, respectively [38]. In addition, the variations of the control inputs are given by

$$\delta\boldsymbol{u}_i = \boldsymbol{u}_i - \boldsymbol{u}_{id} \in \mathbb{R}^3, \quad i = 1, 2. \tag{9}$$

Based on the equations in (8), the linearized dynamics can be constructed using coordinates $(\delta\boldsymbol{p}_0, \boldsymbol{\eta}_0, \boldsymbol{e}_3^{\times}\boldsymbol{\xi}_1, \boldsymbol{e}_3^{\times}\boldsymbol{\xi}_2)$, its derivatives $(\delta\dot{\boldsymbol{p}}_0, \dot{\boldsymbol{\eta}}_0, \boldsymbol{e}_3^{\times}\boldsymbol{e}_3^{\times}\dot{\boldsymbol{\xi}}_1, \boldsymbol{e}_3^{\times}\boldsymbol{e}_3^{\times}\dot{\boldsymbol{\xi}}_2)$, the control inputs $\delta\boldsymbol{u}_i$ $(i = 1, 2)$, and the disturbances $(\boldsymbol{d}_{\boldsymbol{p}_0}, \boldsymbol{d}_{\boldsymbol{R}_0}, \boldsymbol{d}_{\boldsymbol{p}_1}, \boldsymbol{d}_{\boldsymbol{p}_2})$. Since the third element of $\boldsymbol{\xi}_i$ $(i = 1, 2)$ and $\dot{\boldsymbol{\xi}}_i$ $(i = 1, 2)$ is not used in the coordinates and its derivatives, the matrix $\boldsymbol{E}_{12} = [\boldsymbol{e}_1, \boldsymbol{e}_2]^{\top}$ is introduced to simplify the state vector. The final state vector and its derivative are given by

$$\begin{aligned}
\delta\boldsymbol{s} &= [\delta\boldsymbol{p}_0^{\top}, \boldsymbol{\eta}_0^{\top}, \boldsymbol{\xi}_1^{\top}\boldsymbol{E}_{12}^{\top}, \boldsymbol{\xi}_2^{\top}\boldsymbol{E}_{12}^{\top}]^{\top} \in \mathbb{R}^{10},\\
\delta\dot{\boldsymbol{s}} &= [\delta\dot{\boldsymbol{p}}_0^{\top}, \dot{\boldsymbol{\eta}}_0^{\top}, \dot{\boldsymbol{\xi}}_1^{\top}\boldsymbol{E}_{12}^{\top}, \dot{\boldsymbol{\xi}}_2^{\top}\boldsymbol{E}_{12}^{\top}]^{\top} \in \mathbb{R}^{10},
\end{aligned} \tag{10}$$

and the corresponding control inputs and disturbances are given by

$$\begin{aligned}
\delta\boldsymbol{u} &= [\delta\boldsymbol{u}_1^{\top}, \delta\boldsymbol{u}_2^{\top}]^{\top} \in \mathbb{R}^6,\\
\boldsymbol{d} &= [\boldsymbol{d}_{\boldsymbol{p}_0}^{\top}, \boldsymbol{d}_{\boldsymbol{p}_1}^{\top}, \boldsymbol{d}_{\boldsymbol{p}_2}^{\top}, \boldsymbol{d}_{\boldsymbol{R}_0}^{\top}]^{\top} \in \mathbb{R}^{12}.
\end{aligned} \tag{11}$$

The linearized equations of the simplified dynamic model are given in matrix form by

$$\boldsymbol{M}\delta\ddot{\boldsymbol{s}} + \boldsymbol{K}\delta\boldsymbol{s} = \boldsymbol{L}_u\delta\boldsymbol{u} + \boldsymbol{L}_d\boldsymbol{d}, \tag{12}$$

where the explicit expressions of the matrices $\boldsymbol{M}$, $\boldsymbol{K}$, $\boldsymbol{L}_u$ and $\boldsymbol{L}_d$ are given in (13), as shown at the bottom of the next page.

Remark 4: Although linearization based on local coordinates (e.g. rope swing angles in [39]) can also lead to a linear model, these local coordinates may lead to ambiguities of the attitude representations, e.g., ambiguity of Euler angles. In contrast, the variation-based linearization method here gives a unique and coordinate-free treatment [32].

By combining the system kinematics $\delta\dot{\boldsymbol{s}} = \delta\dot{\boldsymbol{s}}$, the linearized system can be written in a state space form as

$$\delta\dot{\bar{\boldsymbol{s}}} = \bar{\boldsymbol{A}}\delta\bar{\boldsymbol{s}} + \bar{\boldsymbol{B}}\delta\boldsymbol{u} + \bar{\boldsymbol{D}}\boldsymbol{d}, \tag{14}$$

where $\delta\bar{\boldsymbol{s}} = [\delta\boldsymbol{s}^{\top}, \delta\dot{\boldsymbol{s}}^{\top}]^{\top} \in \mathbb{R}^{20}$ and

$$\begin{aligned}
&\bar{\boldsymbol{A}} = \begin{bmatrix} \boldsymbol{0}_{10\times 10} & \boldsymbol{I}_{10} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{0}_{10\times 10} \end{bmatrix}, \quad \bar{\boldsymbol{B}} = \begin{bmatrix} \boldsymbol{0}_{10\times 6} \\ \boldsymbol{M}^{-1}\boldsymbol{L}_u \end{bmatrix},\\
&\bar{\boldsymbol{D}} = \begin{bmatrix} \boldsymbol{0}_{10\times 12} \\ \boldsymbol{M}^{-1}\boldsymbol{L}_d \end{bmatrix}.
\end{aligned} \tag{15}$$

The rotation freedom of the pipe $\eta_{0,x}$ is not under control in this modeling setting [33] and is subject to the disturbance $d_{R_0,x}$:

$$\ddot{\eta}_{0,x} = \frac{1}{J_{0,x}}d_{R_0,x}, \tag{16}$$

which can be derived by expanding the equation (14). In practice, this rotation freedom is constrained by the mechanism of the cable attatchment. In addition, the change of $\eta_{0,x}$ has no first-order effects on the other states in (10). Therefore, we choose to neglect the states $\eta_{0,x}$ and $\dot{\eta}_{0,x}$, and assume that this zero dynamics are stable.

Omitting the rotation freedom $\eta_{0,x}$ and its derivative $\dot{\eta}_{0,x}$, the reduced state vector and its derivative are selected as

$$\begin{aligned}
\delta\boldsymbol{s}^r &= [\delta\boldsymbol{p}_0^{\top}, \boldsymbol{\eta}_0^{\top}\boldsymbol{E}_{23}^{\top}, \boldsymbol{\xi}_1^{\top}\boldsymbol{E}_{12}^{\top}, \boldsymbol{\xi}_2^{\top}\boldsymbol{E}_{12}^{\top}]^{\top} \in \mathbb{R}^9,\\
\delta\dot{\boldsymbol{s}}^r &= [\delta\dot{\boldsymbol{p}}_0^{\top}, \dot{\boldsymbol{\eta}}_0^{\top}\boldsymbol{E}_{23}^{\top}, \dot{\boldsymbol{\xi}}_1^{\top}\boldsymbol{E}_{12}^{\top}, \dot{\boldsymbol{\xi}}_2^{\top}\boldsymbol{E}_{12}^{\top}]^{\top} \in \mathbb{R}^9,
\end{aligned} \tag{17}$$

where the matrix $\boldsymbol{E}_{23} = [\boldsymbol{e}_2, \boldsymbol{e}_3]^{\top}$ is introduced. Under Assumption 1, the reduced linearized equations of the simplified model are given in the matrix form by

$$\boldsymbol{M}^r\delta\ddot{\boldsymbol{s}}^r + \boldsymbol{K}^r\delta\boldsymbol{s}^r = \boldsymbol{L}_u^r\delta\boldsymbol{u} + \boldsymbol{L}_d^r\boldsymbol{d}, \tag{18}$$

[1]States and inputs of the desired equilibrium are represented with a subscript d.

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where the explicit expressions of the matrices $\boldsymbol{M}^r$, $\boldsymbol{K}^r$, $\boldsymbol{L}_u^r$ and $\boldsymbol{L}_d^r$ are given in (19), as shown at the bottom of the page.

By combining the system kinematics $\delta\dot{\boldsymbol{s}}^r = \delta\dot{\bar{\boldsymbol{s}}}^r$, the reduced linearized system can be rewritten as

$$\delta\dot{\bar{\boldsymbol{s}}}^r = \bar{\boldsymbol{A}}^r\delta\bar{\boldsymbol{s}}^r + \bar{\boldsymbol{B}}^r\delta\boldsymbol{u} + \bar{\boldsymbol{D}}^r\boldsymbol{d}, \tag{20}$$

where $\delta\bar{\boldsymbol{s}}^r = [(\delta\boldsymbol{s}^r)^\top, (\delta\dot{\boldsymbol{s}}^r)^\top]^\top \in \mathbb{R}^{18}$ and

$$\bar{\boldsymbol{A}}^r = \begin{bmatrix} \boldsymbol{0}_{9\times 9} & \boldsymbol{I}_9 \\ -(\boldsymbol{M}^r)^{-1}\boldsymbol{K}^r & \boldsymbol{0}_{9\times 9} \end{bmatrix}, \quad \bar{\boldsymbol{B}}^r = \begin{bmatrix} \boldsymbol{0}_{9\times 6} \\ (\boldsymbol{M}^r)^{-1}\boldsymbol{L}_u^r \end{bmatrix},$$
$$\bar{\boldsymbol{D}}^r = \begin{bmatrix} \boldsymbol{0}_{9\times 12} \\ (\boldsymbol{M}^r)^{-1}\boldsymbol{L}_d^r \end{bmatrix}. \tag{21}$$

B. Coordinate Transformation

We assume that only the states of the quadrotors $(\boldsymbol{p}_i, \dot{\boldsymbol{p}}_i, \boldsymbol{R}_i, \boldsymbol{\Omega}_i)$ $(i = 1, 2)$ are available for feedback control, which can be seen as outputs of the complete nonlinear model composed by (3), (4), and (5). Similar to the common practice which changes the coordinates of the system from the configuration space to the operation space, we can change the system coordinates by utilizing the holonomic constraints (1). Linearizing the equation (1) yields

$$\begin{aligned} \delta\boldsymbol{p}_i &= \delta\boldsymbol{p}_0 + \delta\boldsymbol{R}_0\boldsymbol{\rho}_i - l_i\delta\boldsymbol{q}_i \\ &= \delta\boldsymbol{p}_0 - \boldsymbol{\rho}_i^\times\boldsymbol{\eta}_0 + l_i\boldsymbol{e}_3^\times\boldsymbol{\xi}_i \\ &= \delta\boldsymbol{p}_0 - \boldsymbol{\rho}_i^\times\boldsymbol{E}_{23}^\top\boldsymbol{E}_{23}\boldsymbol{\eta}_0 + l_i\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top\boldsymbol{E}_{12}\boldsymbol{\xi}_i, \quad i = 1, 2, \end{aligned} \tag{22}$$

where $\delta\boldsymbol{p}_i = \boldsymbol{p}_i - \boldsymbol{p}_{id}$ $(i = 1, 2)$ denotes the infinitesimal variation of the CoM position of the ith quadrotor, and $\boldsymbol{\rho}_i$ $(i = 1, 2)$ denotes the vector pointing from the CoM of the payload to the ith cable attachment point on the payload with respect to the payload frame $\mathcal{F}^{\mathcal{B}_0}$, as mentioned in (2).

Then, the new state vector and its derivative are defined as follows

$$\begin{aligned} \delta\boldsymbol{s}^{ct} &= [\delta\boldsymbol{p}_1^\top, \delta\boldsymbol{p}_2^\top, \boldsymbol{\xi}_1^\top\boldsymbol{E}_{12}^\top, \boldsymbol{\xi}_2^\top\boldsymbol{E}_{12}^\top]^\top \in \mathbb{R}^{10}, \\ \delta\dot{\boldsymbol{s}}^{ct} &= [\delta\dot{\boldsymbol{p}}_1^\top, \delta\dot{\boldsymbol{p}}_2^\top, \dot{\boldsymbol{\xi}}_1^\top\boldsymbol{E}_{12}^\top, \dot{\boldsymbol{\xi}}_2^\top\boldsymbol{E}_{12}^\top]^\top \in \mathbb{R}^{10}, \end{aligned} \tag{23}$$

satisfying

$$\delta\boldsymbol{s}^{ct} = \boldsymbol{T}\delta\boldsymbol{s}^r, \tag{24}$$

$$\boldsymbol{M} = \begin{bmatrix} m_T\boldsymbol{I}_3 & -\sum_{i=1}^2 m_i\boldsymbol{\rho}_i^\times & m_1l_1\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top & m_2l_2\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top \\ \sum_{i=1}^2 m_i\boldsymbol{\rho}_i^\times & \bar{\boldsymbol{J}}_0 & m_1l_1\boldsymbol{\rho}_1^\times\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top & m_2l_2\boldsymbol{\rho}_2^\times\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top \\ -m_1\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & m_1\boldsymbol{E}_{12}\boldsymbol{e}_3^\times\boldsymbol{\rho}_1^\times & m_1l_1\boldsymbol{I}_2 & \boldsymbol{0}_{2\times 2} \\ -m_2\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & m_2\boldsymbol{E}_{12}\boldsymbol{e}_3^\times\boldsymbol{\rho}_2^\times & \boldsymbol{0}_{2\times 2} & m_2l_2\boldsymbol{I}_2 \end{bmatrix}, \quad \boldsymbol{K} = \begin{bmatrix} \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & \frac{m_0g\rho_2}{\rho_1+\rho_2}\boldsymbol{I}_2 & \boldsymbol{0}_{2\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \frac{m_0g\rho_1}{\rho_1+\rho_2}\boldsymbol{I}_2 \end{bmatrix},$$
$$\boldsymbol{L}_u = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{I}_3 \\ \boldsymbol{\rho}_1^\times & \boldsymbol{\rho}_2^\times \\ -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times \end{bmatrix}, \quad \boldsymbol{L}_d = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{I}_3 & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times 3} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{\rho}_1^\times & \boldsymbol{\rho}_2^\times & \boldsymbol{I}_3 \\ \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} \end{bmatrix} \tag{13}$$

$$\boldsymbol{M}^r = \begin{bmatrix} m_T\boldsymbol{I}_3 & -\sum_{i=1}^2 m_i\boldsymbol{\rho}_i^\times\boldsymbol{E}_{23}^\top & m_1l_1\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top & m_2l_2\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top \\ \sum_{i=1}^2 m_i\boldsymbol{E}_{23}\boldsymbol{\rho}_i^\times & \boldsymbol{E}_{23}\bar{\boldsymbol{J}}_0\boldsymbol{E}_{23}^\top & m_1l_1\boldsymbol{E}_{23}\boldsymbol{\rho}_1^\times\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top & m_2l_2\boldsymbol{E}_{23}\boldsymbol{\rho}_2^\times\boldsymbol{e}_3^\times\boldsymbol{E}_{12}^\top \\ -m_1\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & m_1\boldsymbol{E}_{12}\boldsymbol{e}_3^\times\boldsymbol{\rho}_1^\times\boldsymbol{E}_{23}^\top & m_1l_1\boldsymbol{I}_2 & \boldsymbol{0}_{2\times 2} \\ -m_2\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & m_2\boldsymbol{E}_{12}\boldsymbol{e}_3^\times\boldsymbol{\rho}_2^\times\boldsymbol{E}_{23}^\top & \boldsymbol{0}_{2\times 2} & m_2l_2\boldsymbol{I}_2 \end{bmatrix},$$
$$\boldsymbol{K}^r = \begin{bmatrix} \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \frac{m_0g\rho_2}{\rho_1+\rho_2}\boldsymbol{I}_2 & \boldsymbol{0}_{2\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \frac{m_0g\rho_1}{\rho_1+\rho_2}\boldsymbol{I}_2 \end{bmatrix}, \quad \boldsymbol{L}_u^r = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{I}_3 \\ \boldsymbol{E}_{23}\boldsymbol{\rho}_1^\times & \boldsymbol{E}_{23}\boldsymbol{\rho}_2^\times \\ -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times \end{bmatrix},$$
$$\boldsymbol{L}_d^r = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{I}_3 & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{E}_{23}\boldsymbol{\rho}_1^\times & \boldsymbol{E}_{23}\boldsymbol{\rho}_2^\times & \boldsymbol{E}_{23} \\ \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 3} & -\boldsymbol{E}_{12}\boldsymbol{e}_3^\times & \boldsymbol{0}_{2\times 3} \end{bmatrix} \tag{19}$$

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where the matrix $\boldsymbol{T} \in \mathbb{R}^{10\times 9}$ is

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{I}_3 & -\boldsymbol{\rho}_1^{\times}\boldsymbol{E}_{23}^{\top} & l_1\boldsymbol{e}_3^{\times}\boldsymbol{E}_{12}^{\top} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{I}_3 & -\boldsymbol{\rho}_2^{\times}\boldsymbol{E}_{23}^{\top} & \boldsymbol{0}_{3\times 2} & l_2\boldsymbol{e}_3^{\times}\boldsymbol{E}_{12}^{\top} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \boldsymbol{I}_2 & \boldsymbol{0}_{2\times 2} \\ \boldsymbol{0}_{2\times 3} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \boldsymbol{I}_2 \end{bmatrix}. \tag{25}$$

The matrix $\boldsymbol{T}$ is of full column rank so that this matrix can be seen as an immersion mapping and has a left inverse $\boldsymbol{T}^{\dagger} \in \mathbb{R}^{9\times 10}$. Therefore, the dynamic model (20) can be transformed into

$$\delta\dot{\bar{\boldsymbol{s}}}^{ct} = \bar{\boldsymbol{A}}^{ct}\delta\bar{\boldsymbol{s}}^{ct} + \bar{\boldsymbol{B}}^{ct}\delta\boldsymbol{u} + \bar{\boldsymbol{D}}^{ct}\boldsymbol{d}, \tag{26}$$

where $\delta\bar{\boldsymbol{s}}^{ct} = [(\delta\boldsymbol{s}^{ct})^{\top}, (\delta\dot{\boldsymbol{s}}^{ct})^{\top}]^{\top} \in \mathbb{R}^{20}$ and

$$\bar{\boldsymbol{A}}^{ct} = \begin{bmatrix} \boldsymbol{0}_{10\times 10} & \boldsymbol{I}_{10} \\ -\boldsymbol{T}(\boldsymbol{M}^r)^{-1}\boldsymbol{K}^r\boldsymbol{T}^{\dagger} & \boldsymbol{0}_{10\times 10} \end{bmatrix},$$
$$\bar{\boldsymbol{B}}^{ct} = \begin{bmatrix} \boldsymbol{0}_{10\times 6} \\ \boldsymbol{T}(\boldsymbol{M}^r)^{-1}\boldsymbol{L}_u^r \end{bmatrix}, \quad \bar{\boldsymbol{D}}^{ct} = \begin{bmatrix} \boldsymbol{0}_{10\times 12} \\ \boldsymbol{T}(\boldsymbol{M}^r)^{-1}\boldsymbol{L}_d^r \end{bmatrix}. \tag{27}$$

C. Linearized Dynamics of Out-of-Plane Oscillation

We consider the oscillation of the payload in the Y direction, or the out-of-plane dynamics as shown in Fig. 2, which can be extracted from the linear model (26) as below

$$\begin{aligned}
\delta\ddot{p}_{1,y} &= -\frac{m_0 g\rho_2}{m_1(\rho_1+\rho_2)}\xi_{1,x} + \frac{1}{m_1}\delta u_{1,y} + \frac{1}{m_1}d_{p_1,y}, \\
\delta\ddot{p}_{2,y} &= -\frac{m_0 g\rho_1}{m_2(\rho_1+\rho_2)}\xi_{2,x} + \frac{1}{m_2}\delta u_{2,y} + \frac{1}{m_2}d_{p_2,y}, \\
\ddot{\xi}_{1,x} &= -\frac{g\rho_2(m_0 m_1\rho_1^2 + J_{0,z}m_0 + J_{0,z}m_1)}{J_{0,z}m_1 l_1(\rho_1+\rho_2)}\xi_{1,x} \\
&\quad + \frac{1}{m_1 l_1}\delta u_{1,y} - \frac{g\rho_1(J_{0,z} - m_0\rho_1\rho_2)}{J_{0,z}l_1(\rho_1+\rho_2)}\xi_{2,x} \\
&\quad + \frac{1}{m_1 l_1}d_{p_1,y} - \frac{1}{m_0 l_1}d_{p_0,y} - \frac{\rho_1}{J_{0,z}l_1}d_{R_0,z}, \\
\ddot{\xi}_{2,x} &= -\frac{g\rho_1(m_0 m_2\rho_2^2 + J_{0,z}m_0 + J_{0,z}m_2)}{J_{0,z}m_2 l_2(\rho_1+\rho_2)}\xi_{2,x} \\
&\quad + \frac{1}{m_2 l_2}\delta u_{2,y} - \frac{g\rho_2(J_{0,z} - m_0\rho_1\rho_2)}{J_{0,z}l_2(\rho_1+\rho_2)}\xi_{1,x} \\
&\quad + \frac{1}{m_2 l_2}d_{p_2,y} - \frac{1}{m_0 l_2}d_{p_0,y} + \frac{\rho_2}{J_{0,z}l_2}d_{R_0,z}.
\end{aligned} \tag{28}$$

Remark 5: A similar method of modeling and control can also be applied to the payload oscillation in other directions, but we do not expand it here due to space limitation.

Remark 6: In [31], the dynamics of the cable and the payload are not considered, and the cable force on the quadrotor is modeled as a type of periodic disturbance independent of the quadrotor dynamics. In contrast, the proposed model (28) provides a more detailed and precise description of the transportation system, which reveals the relationship between the states of the quadrotors and the states of the cables clearly.

Assumption 1: The mass distribution of the pipe is assumed to be uniform so that the CoM of the payload is located at the center point of the line connecting the two suspension points, i.e., $\rho_1 = \rho_2 = l_0$ or $\boldsymbol{\rho}_1 + \boldsymbol{\rho}_2 = \boldsymbol{0}_{3\times 1}$, where l_0 is the half length of the pipe. Under this condition, the moment of inertia about the body-attached axis $\boldsymbol{z}^{\mathcal{B}_0}$ satisfies $J_{0,z} = \frac{1}{3}m_0 l_0^2$.

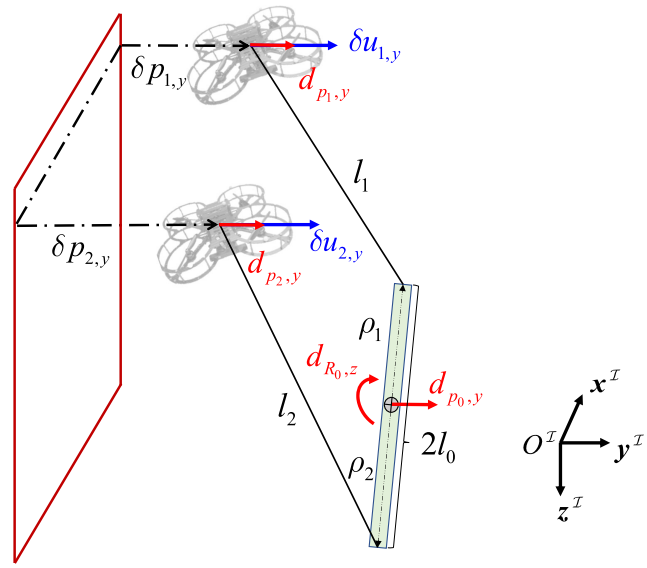

Fig. 2. Out-of-plane oscillation of the payload.

To transform the system (28) into a cascade structure for control purpose, we introduce the coordinate transformation

$$\zeta_i = \xi_{i,x} - \frac{1}{l_i}\delta p_{i,y}, \quad i = 1, 2, \tag{29}$$

where $-l_i\zeta_i = \delta p_{i,y} - l_i\xi_{i,x}$ $(i = 1, 2)$ corresponds to the variation of Y position of the ith attachment point on the payload. By combining Assumption 1 and the coordinate transformation (29), the model (28) can be simplified as

$$\begin{aligned}
\delta\ddot{p}_{i,y} &= -\frac{m_0 g}{2m_i l_i}\delta p_{i,y} - \frac{m_0 g}{2m_i}\zeta_i + \frac{1}{m_i}\delta u_{i,y} + \frac{1}{m_i}d_{p_i,y}, \\
\ddot{\zeta}_i &= -\frac{2g}{l_i}\zeta_i + \frac{g}{l_i}\zeta_j - \frac{2g}{l_i^2}\delta p_{i,y} + \frac{g}{l_1 l_2}\delta p_{j,y} - \frac{1}{m_0 l_i}d_{p_0,y} \\
&\quad + \frac{(-1)^i\cdot 3}{m_0 l_0 l_i}d_{R_0,z}, \quad i, j = 1, 2, \quad i \neq j.
\end{aligned} \tag{30}$$

Remark 7: Assumption 1 describes a specific condition that the mass distribution of the pipe is uniform, which is a common situation for industrial products. This condition is utilized to simplify the expressions in the rest of the paper, but it does not affect the generality of the proposed method. To be explicit, for the mass-center-offset or nonuniform-mass-distribution case, the coordinate transformation in (29) can also be utilized to simplify the model (28), which would result in a more complicated model than the current model (30), and if the model parameters ρ_1, ρ_2, and $J_{0,z}$ are known, similar procedures in section IV-A and IV-B can be applied for cooperative estimation and control of the two-drone-payload system.

From the perspective of anti-disturbance control [40], [41], [42], [43], the equations in (30) can be further split into two parts. The first part is the model of the quadrotors as

$$\dot{\boldsymbol{x}}_i = \boldsymbol{A}_i\boldsymbol{x}_i + \boldsymbol{B}_i(\delta u_{i,y} + d_{\zeta_i} + d_{p_i,y}), \quad i = 1, 2, \tag{31}$$

where $\boldsymbol{x}_i = [\delta p_{i,y}, \delta\dot{p}_{i,y}]^{\top}$ $(i = 1, 2)$ corresponds to the states of the quadrotors, $d_{\zeta_i} = -\frac{m_0 g}{2}\zeta_i$ $(i = 1, 2)$ and $d_{p_i,y}$ $(i = 1, 2)$

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can be treated as the matched disturbance, and

$$\boldsymbol{A}_i = \begin{bmatrix} 0 & 1 \\ -\dfrac{m_0 g}{2m_i l_i} & 0 \end{bmatrix}, \quad \boldsymbol{B}_i = \begin{bmatrix} 0 \\ \dfrac{1}{m_i} \end{bmatrix}, \quad i = 1, 2.$$

The second part is the exogenous model of the disturbance d_{ζ_i} as

$$\begin{aligned} \dot{\boldsymbol{w}}_i &= \boldsymbol{W}_{ii}\boldsymbol{w}_i + \boldsymbol{W}_{ij}\boldsymbol{w}_j + \boldsymbol{H}_{ii}\boldsymbol{x}_i + \boldsymbol{H}_{ij}\boldsymbol{x}_j + \boldsymbol{D}_i\boldsymbol{d}_0, \\ d_{\zeta_i} &= \boldsymbol{V}\boldsymbol{w}_i, \quad i, j = 1, 2, \quad i \neq j, \end{aligned} \tag{32}$$

where the internal state of the disturbance d_{ζ_i} is denoted by

$$\boldsymbol{w}_i = [\zeta_i, \dot{\zeta}_i]^\top, \quad i = 1, 2, \tag{33}$$

and the unmodeled disturbance is denoted by

$$\boldsymbol{d}_0 = [d_{p_0,y}, d_{R_0,z}]^\top. \tag{34}$$

The matrices used in (32) are listed below as

$$\boldsymbol{W}_{ii} = \begin{bmatrix} 0 & 1 \\ -\dfrac{2g}{l_i} & 0 \end{bmatrix}, \boldsymbol{W}_{ij} = \begin{bmatrix} 0 & 0 \\ \dfrac{g}{l_i} & 0 \end{bmatrix}, \quad \boldsymbol{H}_{ii} = \begin{bmatrix} 0 & 0 \\ -\dfrac{2g}{l_i^2} & 0 \end{bmatrix},$$

$$\boldsymbol{H}_{ij} = \begin{bmatrix} 0 & 0 \\ \dfrac{g}{l_1 l_2} & 0 \end{bmatrix}, \quad \boldsymbol{D}_i = \begin{bmatrix} 0 & 0 \\ -\dfrac{1}{m_0 l_i} & \dfrac{(-1)^i \cdot 3}{m_0 l_0 l_i} \end{bmatrix},$$

$$\boldsymbol{V} = \begin{bmatrix} -\dfrac{m_0 g}{2} & 0 \end{bmatrix}, \quad i, j = 1, 2, \quad i \neq j.$$

Assumption 2: There exist unknown upper bounds for $d_{p_0,y}$, $d_{R_0,z}$, and $\dot{d}_{p_i,y}$ $(i = 1, 2)$, i.e.,

$$\begin{aligned} |d_{p_0,y}| &\leq \bar{d}_1, \\ |d_{R_0,z}| &\leq \bar{d}_2, \\ |\dot{d}_{p_i,y}| &\leq \bar{d}_3, \quad i = 1, 2. \end{aligned} \tag{35}$$

Remark 8: The disturbance force $d_{p_0,y}$ and the disturbance torque $d_{R_0,z}$ on the payload are assumed to be upper bounded, which is reasonable since the changing rate of the practical physical systems is inherently bounded. In addition, the disturbance force $d_{p_i,y}$ $(i = 1, 2)$ on the ith quadrotor, including mass uncertainty and motor degradation, changes slowly around its equilibrium so that the time derivative of $d_{p_i,y}$ $(i = 1, 2)$ can be assumed upper bounded [31].

The coupling between the states of the disturbance $\boldsymbol{w}_i$ $(i = 1, 2)$ and the states of the quadrotors $\boldsymbol{x}_i$ $(i = 1, 2)$ is explicitly presented in the model (32), which illustrates that the established model is more refined than the one used in [31]. Since the methods in [41], [42], and [43] are developed only for estimation of the disturbance independent of the system states, they are not applicable for estimation of this coupled disturbance. Hence, an effective solution to estimating the disturbance d_{ζ_i} and $d_{p_i,y}$ separately is proposed in the next section.

IV. Cooperative Control Algorithm Design

In this section, a refined cooperative disturbance estimation strategy is designed to estimate the modeled disturbance d_{ζ_i} and the derivative-bounded disturbance $d_{p_i,y}$, separately. Based on the estimates, cooperative controllers are then developed to stabilize the whole system.

A. Refined Cooperative Disturbance Estimation Strategy

For the aerial co-transportation system considered in this paper, the cable direction $\boldsymbol{q}_i$ $(i = 1, 2)$ is unmeasurable, so the state of the modeled disturbance $\boldsymbol{w}_i$ $(i = 1, 2)$ in (32) can not be obtained directly. Let $\hat{\boldsymbol{w}}_i = [\hat{\zeta}_i, \dot{\hat{\zeta}}_i]^\top$ $(i = 1, 2)$ denote the estimates of $\boldsymbol{w}_i$ $(i = 1, 2)$. Swing observers are designed to estimate $\hat{\boldsymbol{w}}_i$ $(i = 1, 2)$:

$$\begin{aligned} \dot{\boldsymbol{z}}_{\boldsymbol{w}_i} &= (\boldsymbol{W}_{ii} - \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{B}_i\boldsymbol{V})(\boldsymbol{z}_{\boldsymbol{w}_i} + \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{x}_i) + \boldsymbol{W}_{ij}(\boldsymbol{z}_{\boldsymbol{w}_j} + \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{x}_j) \\ &\quad + (\boldsymbol{H}_{ii} - \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{A}_i)\boldsymbol{x}_i + \boldsymbol{H}_{ij}\boldsymbol{x}_j - \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{B}_i(\delta u_{i,y} + \hat{d}_{p_i,y}), \\ \hat{\boldsymbol{w}}_i &= \boldsymbol{z}_{\boldsymbol{w}_i} + \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{x}_i, \\ \hat{d}_{\zeta_i} &= \boldsymbol{V}\hat{\boldsymbol{w}}_i, \quad i, j = 1, 2, \quad i \neq j, \end{aligned} \tag{36}$$

where $\boldsymbol{z}_{\boldsymbol{w}_i}$ $(i = 1, 2)$ is an auxiliary variable, $\boldsymbol{L}_{\boldsymbol{w}} = \mathrm{diag}\{l_{w_1}, l_{w_2}\} \in \mathbb{R}^{2\times 2}$ is the observer gain, $\hat{d}_{\zeta_i} = -\frac{m_0 g}{2}\hat{\zeta}_i$ denotes the estimate of d_{ζ_i}, and $\hat{d}_{p_i,y}$ denotes the estimate of the disturbance $d_{p_i,y}$, which is introduced to reduce the effect of model uncertainties and nonlinearity on estimation accuracy of $\boldsymbol{w}_i$.

The disturbance observer to estimate the derivative-bounded disturbance $d_{p_i,y}$ $(i = 1, 2)$ is designed as [40]

$$\begin{aligned} \dot{z}_{p_i,y} &= -l_p(z_{p_i,y} + l_p m_i \delta\dot{p}_{i,y}) + \frac{m_0 g l_p}{2}\hat{\zeta}_i \\ &\quad + \frac{m_0 g l_p}{2l_i}\delta p_{i,y} - l_p \delta u_{i,y} \\ \hat{d}_{p_i,y} &= z_{p_i,y} + m_i l_p \delta\dot{p}_{i,y}, \end{aligned} \tag{37}$$

where $z_{p_i,y}$ $(i = 1, 2)$ is an auxiliary variable and l_p is the observer gain.

In principle, the swing observer shown in (36) belongs to a specific class of disturbance observers [40], which are designed for the disturbance with exogenous models, and here it is used to capture the states of the cable directions. The disturbance observer shown in (37) belongs to the class of nonlinear disturbance observers [40], which is used to estimate the derivative-bounded disturbance. This combination of the two types of disturbance observers is similar to the anti-disturbance methods in in [41], [42], and [43]. It is noted that $\hat{d}_{p_i,y}$ requires no information from the other quadrotor, while $\hat{\boldsymbol{w}}_i$ requires information exchange of $\boldsymbol{z}_{\boldsymbol{w}_i}$ and $\boldsymbol{x}_i$ with the other quadrotor. These information exchanges improve the estimation performance. Therefore, we claim "refined" to highlight the accurate disturbance modeling rather than simply treating the cable force as a periodic disturbance as in [31], and "cooperative" to explain the utilization of exchanged information in the proposed swing observer.

B. Cooperative Control Law

The control objective is to develop control laws for the two quadrotors to achieve the following behaviors.

1) The Y positions of quadrotor 1 and quadrotor 2 converge to the desired equilibrium:

$$\boldsymbol{x}_i = [\delta p_{i,y}, \delta\dot{p}_{i,y}]^\top \to \boldsymbol{0}_{2\times 1}, \quad i = 1, 2. \tag{38}$$

2) The Y positions of the attachment points on the payload converge to the desired equilibrium:

$$\boldsymbol{w}_i = [\zeta_i, \dot{\zeta}_i]^\top \to \boldsymbol{0}_{2\times 1}, \quad i = 1, 2. \tag{39}$$

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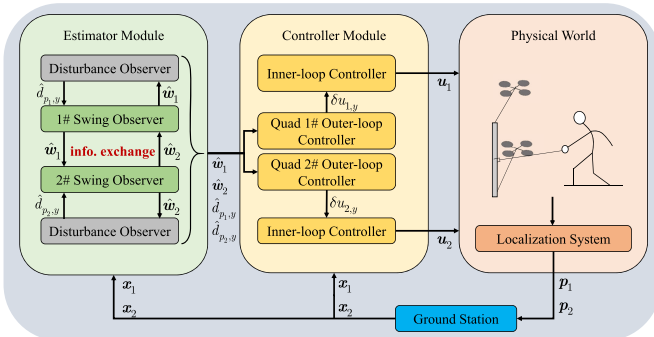

Fig. 3. Control block diagram of the system.

If we directly follow the control design principle in [31], the control input for the ith quadrotor takes the form as

$$\delta u_{i,y} = -\boldsymbol{K_x}\boldsymbol{x}_i - \hat{d}_{\zeta_i} - \hat{d}_{p_i,y}, \quad i = 1, 2, \tag{40}$$

where $\boldsymbol{K_x} = [k_{x_1}, k_{x_2}]$ is the control gain for position stabilization of the quadrotors, $\hat{d}_{\zeta_i}$ $(i = 1, 2)$ and $\hat{d}_{p_i,y}$ $(i = 1, 2)$ are used for disturbance compensation. Since this controller (40) simply compensates all the disturbance encountered by the quadrotors, only precise position tracking of the quadrotors can be achieved.

Different from the control design (40), our proposed control input $\delta u_{i,y}$ $(i = 1, 2)$ takes the following form

$$\begin{aligned}\delta u_{1,y} &= -\boldsymbol{K_x}\boldsymbol{x}_1 - \boldsymbol{K_w}\hat{\boldsymbol{w}}_1 - \boldsymbol{K}_{co}\hat{\boldsymbol{w}}_2 - \hat{d}_{p_1,y},\\ \delta u_{2,y} &= -\boldsymbol{K_x}\boldsymbol{x}_2 - \boldsymbol{K_w}\hat{\boldsymbol{w}}_2 - \boldsymbol{K}_{co}\hat{\boldsymbol{w}}_1 - \hat{d}_{p_2,y},\end{aligned} \tag{41}$$

where $\boldsymbol{K_w} = [k_{w_1}, k_{w_2}]$, and $\boldsymbol{K}_{co} = [k_{co_1}, k_{co_2}]$ are the control gains for swing suppression. In this cooperative control law (41), $\hat{\boldsymbol{w}}_i = [\zeta_i, \dot{\zeta}_i]^\top$ are treated as the position and velocity feedback of the ith attachment point on the payload for oscillation suppression. In addition, including the information exchange term $\hat{\boldsymbol{w}}_2$ in $\delta u_{1,y}$ and the information exchange term $\hat{\boldsymbol{w}}_1$ in $\delta u_{2,y}$ tend to make the motion of the two quadrotors more coordinated.

The structure of the control system, summarized by the set of Eqs. (36), (37), and (41), is illustrated in Fig. 3.

C. Stability and Convergence Analysis

Combining (31) and (41), the closed-loop system is described as

$$\begin{aligned}\dot{\boldsymbol{x}}_i &= (\boldsymbol{A}_i - \boldsymbol{B}_i\boldsymbol{K_x})\boldsymbol{x}_i + \boldsymbol{B}_i(\boldsymbol{V} - \boldsymbol{K_w})\boldsymbol{w}_i - \boldsymbol{B}_i\boldsymbol{K}_{co}\boldsymbol{w}_j\\ &\quad - \boldsymbol{B}_i\boldsymbol{K_w}\tilde{\boldsymbol{w}}_i - \boldsymbol{B}_i\boldsymbol{K}_{co}\tilde{\boldsymbol{w}}_j - \boldsymbol{B}_i\tilde{d}_{p_i,y},\\ i, j &= 1, 2, \quad i \neq j,\end{aligned} \tag{42}$$

where $\tilde{\boldsymbol{w}}_i = \hat{\boldsymbol{w}}_i - \boldsymbol{w}_i$ $(i = 1, 2)$ denotes the estimation error of the swing observer (36) and $\tilde{d}_{p_i,y} = \hat{d}_{p_i,y} - d_{p_i,y}$ $(i = 1, 2)$ denotes the estimation error of the disturbance observer (37).

Based on (31), (32), (34), and (36), the estimation error dynamics of $\boldsymbol{w}_i$ are derived as

$$\begin{aligned}\dot{\tilde{\boldsymbol{w}}}_i &= (\boldsymbol{W}_{ii} - \boldsymbol{L_w}\boldsymbol{B}_i\boldsymbol{V})\tilde{\boldsymbol{w}}_i + \boldsymbol{W}_{ij}\tilde{\boldsymbol{w}}_j - \boldsymbol{L_w}\boldsymbol{B}_i\tilde{d}_{p_i,y}\\ &\quad - \boldsymbol{D}_i\boldsymbol{d}_0, \quad i, j = 1, 2, \quad i \neq j.\end{aligned} \tag{43}$$

Similarly, the estimation error dynamics of $d_{p_i,y}$ are derived by combining (31) and (37) as

$$\begin{aligned}\dot{\tilde{d}}_{p_i,y} &= -l_p\tilde{d}_{p_i,y} + \frac{m_0 g l_p}{2}\tilde{\zeta}_i - \dot{d}_{p_i,y}\\ &= -l_p\tilde{d}_{p_i,y} - l_p\boldsymbol{V}\tilde{\boldsymbol{w}}_i - \dot{d}_{p_i,y}, \quad i = 1, 2.\end{aligned} \tag{44}$$

Finally, the whole system based on (42), (43), and (44) yields the equation (45), as shown at the bottom of the next page, which can be reorganized into the compact form as

$$\begin{aligned}\dot{\boldsymbol{\delta}}_i &= \boldsymbol{\Pi}_{ii}\boldsymbol{\delta}_i + \boldsymbol{\Pi}_{ij}\boldsymbol{\delta}_j + \boldsymbol{\mu}_i,\\ i, j &= 1, 2, \quad i \neq j.\end{aligned} \tag{46}$$

where the explicit expressions of $\boldsymbol{\delta}_i$ $(i = 1, 2)$, $\boldsymbol{\Pi}_{ij}$ $(i, j = 1, 2)$, and $\boldsymbol{\mu}_i$ $(i = 1, 2)$ are presented in (45). Then the overall dynamics can be obtained as

$$\underbrace{\begin{bmatrix}\dot{\boldsymbol{\delta}}_1\\ \dot{\boldsymbol{\delta}}_2\end{bmatrix}}_{\dot{\boldsymbol{\delta}}} = \underbrace{\begin{bmatrix}\boldsymbol{\Pi}_{11} & \boldsymbol{\Pi}_{12}\\ \boldsymbol{\Pi}_{21} & \boldsymbol{\Pi}_{22}\end{bmatrix}}_{\boldsymbol{\Pi}}\underbrace{\begin{bmatrix}\boldsymbol{\delta}_1\\ \boldsymbol{\delta}_2\end{bmatrix}}_{\boldsymbol{\delta}} + \underbrace{\begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}}_{\boldsymbol{\mu}}. \tag{47}$$

Theorem 1: Consider the system (47) and the given positive parameters γ_1, γ_2. Under Assumption 1 and 2, if there exist constant l_p, gain matrices $\boldsymbol{K_x}, \boldsymbol{K_w}, \boldsymbol{K}_{co}$, and $\boldsymbol{L_w}$ satisfying

$$\boldsymbol{\Pi} + \boldsymbol{\Pi}^\top + 2\gamma_1\boldsymbol{I}_{14} < \boldsymbol{0}_{14\times 14}, \tag{48a}$$

$$\gamma_1 - \frac{1}{4\gamma_2} > 0, \tag{48b}$$

then the signals of the overall closed-loop system $(\boldsymbol{x}_i, \boldsymbol{w}_i, \tilde{\boldsymbol{w}}_i, \tilde{d}_{p_i,y})$ $(i = 1, 2)$ are bounded stable.

Proof: Similar to the proof in [41], considering $\boldsymbol{\mu}$ as the input to the system (47), the remaining $\dot{\boldsymbol{\delta}} = \boldsymbol{\Pi}\boldsymbol{\delta}$ is the original unforced system. Using $V(\boldsymbol{\delta}) = \frac{1}{2}\boldsymbol{\delta}^\top\boldsymbol{\delta}$ as a Lyapunov function candidate, the time derivative of $V(\boldsymbol{\delta})$ along the trajectory of $\dot{\boldsymbol{\delta}} = \boldsymbol{\Pi}\boldsymbol{\delta} + \boldsymbol{\mu}$ is

$$\begin{aligned}\dot{V}(\boldsymbol{\delta}) &= \frac{1}{2}\boldsymbol{\delta}^\top(\boldsymbol{\Pi}^\top + \boldsymbol{\Pi})\boldsymbol{\delta} + \boldsymbol{\mu}^\top\boldsymbol{\delta}\\ &= -\gamma_1\|\boldsymbol{\delta}\|^2 + \sum_{i=1}^{2}\boldsymbol{\mu}_i^\top\boldsymbol{\delta}_i\\ &\le -\gamma_1\|\boldsymbol{\delta}\|^2 + \sum_{i=1}^{2}\left(\boldsymbol{w}_i^\top\boldsymbol{D}_i\boldsymbol{d}_0 - \tilde{\boldsymbol{w}}_i^\top\boldsymbol{D}_i\boldsymbol{d}_0 - \tilde{d}_{p_i,y}\dot{d}_{p_i,y}\right)\\ &\le -\gamma_1\|\boldsymbol{\delta}\|^2 + \sum_{i=1}^{2}\Big[\|\boldsymbol{w}_i\|\|\boldsymbol{D}_i\boldsymbol{d}_0\| + \|\tilde{\boldsymbol{w}}_i\|\|\boldsymbol{D}_i\boldsymbol{d}_0\|\\ &\quad + |\tilde{d}_{p_i,y}||\dot{d}_{p_i,y}|\Big]\end{aligned} \tag{49}$$

where the condition (48a) is used.

Based on the definition of the vector $\boldsymbol{\delta}_i$ $(i = 1, 2)$ in (45) and $\boldsymbol{\delta}$ in (47), the following inequalities hold for $i = 1, 2$

$$\begin{aligned}\|\boldsymbol{w}_i\| &\le \|\boldsymbol{\delta}_i\| \le \|\boldsymbol{\delta}\|\\ \|\tilde{\boldsymbol{w}}_i\| &\le \|\boldsymbol{\delta}_i\| \le \|\boldsymbol{\delta}\|\\ |\tilde{d}_{p_i,y}| &\le \|\boldsymbol{\delta}_i\| \le \|\boldsymbol{\delta}\|\end{aligned} \tag{50}$$

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Substituting (50) into (49) yields

$$\begin{aligned}\dot{V}(\boldsymbol{\delta}) &\le -\gamma_1\|\boldsymbol{\delta}\|^2 + \|\boldsymbol{\delta}\|\cdot\sum_{i=1}^{2}\left(2\|\boldsymbol{D}_i\boldsymbol{d}_0\| + |\dot{d}_{p_i,y}|\right)\\ &= -\gamma_1\|\boldsymbol{\delta}\|^2 + \|\boldsymbol{\delta}\|\\ &\quad\cdot\sum_{i=1}^{2}\left(\left\|\left[-\frac{2}{m_0 l_i}d_{p_0,y},\ \frac{(-1)^i\cdot 6}{m_0 l_0 l_i}d_{R_0,z}\right]^\top\right\| + |\dot{d}_{p_i,y}|\right)\\ &\le -\gamma_1\|\boldsymbol{\delta}\|^2 + \|\boldsymbol{\delta}\|\cdot\sum_{i=1}^{2}\left(\frac{2}{m_0 l_i}|d_{p_0,y}| + \frac{6}{m_0 l_0 l_i}|d_{R_0,z}|\right.\\ &\quad\left. + |\dot{d}_{p_i,y}|\right)\\ &\le -\gamma_1\|\boldsymbol{\delta}\|^2 + \|\boldsymbol{\delta}\|\cdot\sum_{i=1}^{2}\left(\frac{2}{m_0 l_i}\bar{d}_1 + \frac{6}{m_0 l_0 l_i}\bar{d}_2 + \bar{d}_3\right)\end{aligned} \tag{51}$$

where Assumption 2 has been used for inequality scaling. Define the constant $\Delta = \sum_{i=1}^{2}\frac{2}{m_0 l_i}\bar{d}_1 + \frac{6}{m_0 l_0 l_i}\bar{d}_2 + \bar{d}_3$, and it follows from Young's inequality that:

$$\|\boldsymbol{\delta}\|\cdot\Delta \le \gamma_2\Delta^2 + \frac{1}{4\gamma_2}\|\boldsymbol{\delta}\|^2 \tag{52}$$

Therefore, the time derivative of Lyapunov function yields

$$\begin{aligned}\dot{V}(\boldsymbol{\delta}) &\le -(\gamma_1 - \frac{1}{4\gamma_2})\|\boldsymbol{\delta}\|^2 + \gamma_2\Delta^2\\ &= -(2\gamma_1 - \frac{1}{2\gamma_2})V(\boldsymbol{\delta}) + \gamma_2\Delta^2\end{aligned} \tag{53}$$

Based on the condition (48b) and Lyapunov boundedness theory [44], we can conclude that the system (47) is uniformly ultimately bounded, implying the boundedness of $\boldsymbol{\delta}$, and thereby the boundedness of the signals $(\boldsymbol{x}_i, \boldsymbol{w}_i, \tilde{\boldsymbol{w}}_i, \tilde{d}_{p_i,y})$ $(i = 1, 2)$, which completes the proof. □

V. Simulation Studies

Simulation studies based on the complete model [33] are carried out to validate the proposed oscillation suppression-enhanced cooperative control strategy. System and control parameters are listed in Table II, where $\boldsymbol{R}_x(\theta)$ denotes the rotation transformation around the $\boldsymbol{x}^{\mathcal{I}}$ axis by the angle θ

$$\boldsymbol{R}_x(\theta) = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\theta & -\sin\theta\\ 0 & \sin\theta & \cos\theta\end{bmatrix}.$$

TABLE II
Parameters Used in the Simulations

System Parameters		
m_i	$i=0$	$0.445,\mathrm{kg}$
	$i=1$	$0.831,\mathrm{kg}$
	$i=2$	$0.832,\mathrm{kg}$
$\boldsymbol{J}_i$	$i=1$	$\mathrm{diag}\{0.003, 0.003, 0.004\},\mathrm{kg}\cdot\mathrm{m}^2$
	$i=2$	$\mathrm{diag}\{0.003, 0.003, 0.004\},\mathrm{kg}\cdot\mathrm{m}^2$
l_i	$i=0$	$1.0,\mathrm{m}$
	$i=1$	$1.2,\mathrm{m}$
	$i=2$	$1.2,\mathrm{m}$
Control Parameters		
k_{x_1}, k_{x_2}		$2, 4$
k_{w_1}, k_{w_2}		$1.2, 0.3$
k_{co_1}, k_{co_2}		$0.5, 0.3$
l_{w_1}, l_{w_2}		$4, 8$
l_p		2
Desired Equilibrium		
$\boldsymbol{p}_{1d}$		$[1, 0, -2]^\top,\mathrm{m}$
$\boldsymbol{p}_{2d}$		$[-1, 0, -2]^\top,\mathrm{m}$
Initial Conditions		
$\boldsymbol{p}_0(0)$		$[0, -0.60, -0.46]^\top,\mathrm{m}$
$\boldsymbol{R}_0(0)$		$\boldsymbol{R}_x(\pi/6)\cdot\boldsymbol{I}_3$
$\boldsymbol{q}_1(0)$		$\boldsymbol{R}_x(\pi/6)\cdot\boldsymbol{e}_3$
$\boldsymbol{q}_2(0)$		$\boldsymbol{R}_x(\pi/6)\cdot\boldsymbol{e}_3$
$\dot{\boldsymbol{p}}_0$		$[0, 0, 0]^\top,\mathrm{m}\cdot\mathrm{s}^{-1}$
$\boldsymbol{\Omega}_0(0)$		$[0, 0, 0]^\top,\mathrm{rad}\cdot\mathrm{s}^{-1}$
$\boldsymbol{\omega}_1(0)$		$[0, 0, 0]^\top$
$\boldsymbol{\omega}_2(0)$		$[0, 0, 0]^\top$

The disturbance exerted on the quadrotors and the payload are set as

$$\begin{aligned}\boldsymbol{d}_{\boldsymbol{p}_0} &= \begin{cases}[0, 0, 0]^\top,\mathrm{N}, & 0 \le t < 10\mathrm{s}\\ [0, 5, 0]^\top,\mathrm{N}, & 10 \le t < 10.2\mathrm{s}\\ [0, 0, 0]^\top,\mathrm{N}, & 10.2 \le t \le 30\mathrm{s}\end{cases}\\ \boldsymbol{d}_{\boldsymbol{p}_i} &= [0, 0.1, 0]^\top,\mathrm{N}, \quad i = 1, 2\\ \boldsymbol{d}_{\boldsymbol{R}_0} &= \boldsymbol{d}_{\boldsymbol{R}_1} = \boldsymbol{d}_{\boldsymbol{R}_2} = \boldsymbol{0}_{3\times 1},\ \mathrm{N}\cdot\mathrm{m}\end{aligned} \tag{54}$$

In the simulation, the pipe is released from its initial condition at time $t = 0$ and then subject to a sudden impact at $t = 10\mathrm{s}$ as shown in Eq. (54). The control objective is to suppress the payload swing by the two quadrotors for system

$$\underbrace{\begin{bmatrix}\dot{\boldsymbol{x}}_i\\ \dot{\boldsymbol{w}}_i\\ \dot{\tilde{\boldsymbol{w}}}_i\\ \dot{\tilde{d}}_{p_i,y}\end{bmatrix}}_{\dot{\boldsymbol{\delta}}_i} = \underbrace{\begin{bmatrix}\boldsymbol{A}_i - \boldsymbol{B}_i\boldsymbol{K}_{\boldsymbol{x}} & \boldsymbol{B}_i(\boldsymbol{K}_{\boldsymbol{w}} + \boldsymbol{V}) & \boldsymbol{B}_i\boldsymbol{K}_{\boldsymbol{w}} & -\boldsymbol{B}_i\\ \boldsymbol{H}_{ii} & \boldsymbol{W}_{ii} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 1}\\ \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \boldsymbol{W}_{ii} - \boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{B}_i\boldsymbol{V} & -\boldsymbol{L}_{\boldsymbol{w}}\boldsymbol{B}_i\\ \boldsymbol{0}_{1\times 2} & \boldsymbol{0}_{1\times 2} & -l_p\boldsymbol{V} & -l_p\end{bmatrix}}_{\boldsymbol{\Pi}_{ii}}\underbrace{\begin{bmatrix}\boldsymbol{x}_i\\ \boldsymbol{w}_i\\ \tilde{\boldsymbol{w}}_i\\ \tilde{d}_{p_i,y}\end{bmatrix}}_{\boldsymbol{\delta}_i} + \underbrace{\begin{bmatrix}\boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \boldsymbol{B}_i\boldsymbol{K}_{co} & \boldsymbol{0}_{2\times 1}\\ \boldsymbol{H}_{ij} & \boldsymbol{W}_{ij} & \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 1}\\ \boldsymbol{0}_{2\times 2} & \boldsymbol{0}_{2\times 2} & \boldsymbol{W}_{ij} & \boldsymbol{0}_{2\times 1}\\ \boldsymbol{0}_{1\times 2} & \boldsymbol{0}_{1\times 2} & \boldsymbol{0}_{1\times 2} & 0\end{bmatrix}}_{\boldsymbol{\Pi}_{ij}}\underbrace{\begin{bmatrix}\boldsymbol{x}_j\\ \boldsymbol{w}_j\\ \tilde{\boldsymbol{w}}_j\\ \tilde{d}_{p_j,y}\end{bmatrix}}_{\boldsymbol{\delta}_j}$$

$$+ \underbrace{\begin{bmatrix}\boldsymbol{0}_{2\times 1}\\ \boldsymbol{D}_i\boldsymbol{d}_0\\ -\boldsymbol{D}_i\boldsymbol{d}_0\\ -\dot{d}_{p_i,y}\end{bmatrix}}_{\boldsymbol{\mu}_i}, \quad i, j = 1, 2, \quad i \ne j \tag{45}$$

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stabilization. The desired positions $\boldsymbol{p}_{id}$ $(i = 1, 2)$ of the two quadrotors are chosen based on the desired equilibrium of the pipe $(\boldsymbol{p}_{0d}, \boldsymbol{R}_{0d})$, the length of the pipe $2l_0$, and the length of the ith cable l_i $(i = 1, 2)$.

In addition, five methods (NHC in [21], IDA-PBC in [22], MOBADC in [31], PO in [23], FT in [24]) are chosen for comparison since they also only use the states of the quadrotors for stabilization of the whole system.

A. Attachment Point Estimation

First, the positions of the attachment points on the payload with respect to the inertial frame $\mathcal{F}^{\mathcal{I}}$ can be approximately estimated based on the definition of the vector $\boldsymbol{\xi}_i$ $(i = 1, 2)$ in (8) as

$$\begin{aligned} \boldsymbol{p}_{0i} &= \boldsymbol{p}_i + l_i \boldsymbol{q}_i = \delta \boldsymbol{p}_i + l_i \delta \boldsymbol{q}_i + \boldsymbol{p}_{id} + \boldsymbol{l}_i \boldsymbol{q}_{id} \\ &\approx \delta \boldsymbol{p}_i - l_i \boldsymbol{e}_3^{\times} \boldsymbol{\xi}_i + \boldsymbol{p}_{id} + \boldsymbol{l}_i \boldsymbol{q}_{id}, \\ i &= 1, 2. \end{aligned} \tag{55}$$

Then, the Y position of the ith attachment point on the payload can be estimated by

$$\hat{p}_{0i,y} = \delta p_{i,y} - l_i \hat{\xi}_{i,x} = -l_i \hat{\zeta}_i, \quad i = 1, 2, \tag{56}$$

where $\hat{\xi}_{i,x}$ $(i = 1, 2)$ and $\hat{\zeta}_i$ $(i = 1, 2)$ denote the estimates of the states $\xi_{i,x}$ $(i = 1, 2)$ and ζ_i $(i = 1, 2)$, respectively.

To give out the ground truth, the positions of the attachment points on the payload $\boldsymbol{p}_{0i} \in \mathbb{R}^3$ $(i = 1, 2)$ are computed by

$$\boldsymbol{p}_{0i} = \boldsymbol{p}_0 + (-1)^{i+1} l_0 \boldsymbol{R}_0 \boldsymbol{e}_1, \quad i = 1, 2. \tag{57}$$

Here, the Y positions of the attachment points are expanded as

$$p_{0i,y} = p_{0,y} + (-1)^{i+1} l_0 \cos\theta_0 \sin\psi_0, \quad i = 1, 2, \tag{58}$$

where the Euler angles θ_0 and ψ_0 representing the attitude of the payload. Therefore, the expressions (58) can be used to compute the ground truth of the Y positions $p_{0i,y}$ $(i = 1, 2)$.

The comparison of the actual states $p_{0i,y}$ $(i = 1, 2)$ and the estimates $\hat{p}_{0i,y}$ $(i = 1, 2)$ in the simulation is depicted in Fig. 4. It can be concluded from Fig. 4 that whether the system is subject to the initial state offset at $t = 0$ or external disturbance excitation at $t = 10$s, the linearized estimates can quickly converge to the actual states in a short time period so that they can be utilized for real time feedback control.

B. Swing Suppression

Time evolution of the payload position in the Y direction for different methods is depicted in Fig. 5, from which an intuitive observation is that all the methods can successfully suppress the payload oscillation, and the proposed method achieves the fastest convergence speed in payload swing suppression whether under the initial state offset at $t = 0$ or external disturbance excitation at $t = 10$s in the simulation. This result reveals the superiority of the proposed method in oscillation suppression of the payload over the other methods.

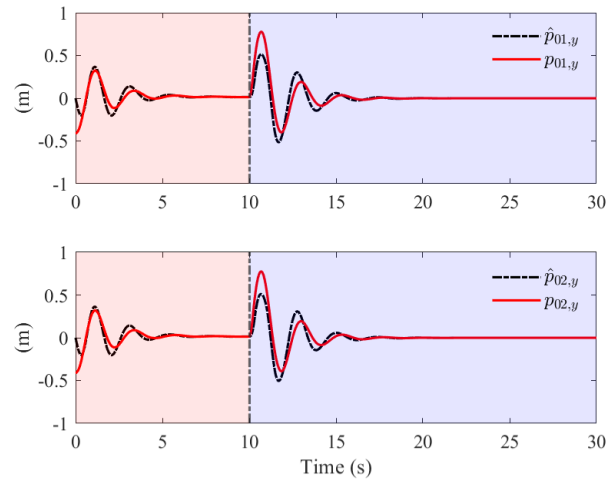

Fig. 4. Estimation results of the Y positions of the attachment points in the simulation. The light red part denotes the time period of response to the initial state offset at $t = 0$, and the light blue part denotes the time period of response to the sudden impact on the payload at $t = 10$s.

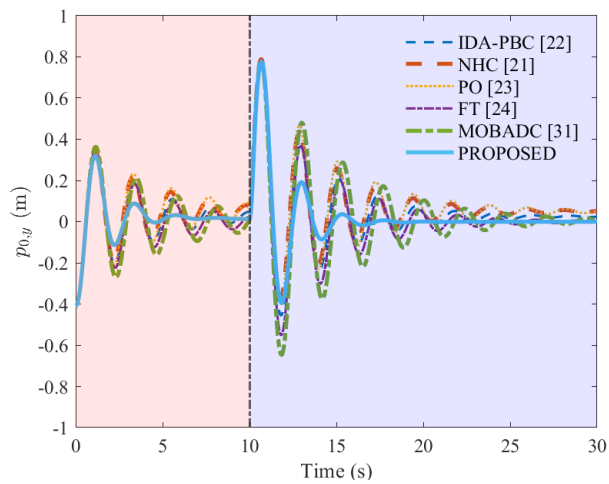

Fig. 5. Time evolution of the CoM position of the payload in the Y direction in the simulation. The light red part denotes the time period of response to the initial state offset at $t = 0$, and the light blue part denotes the time period of response to the sudden impact on the payload at $t = 10$s.

VI. Experimental Implementation

To demonstrate the effectiveness of the proposed refined cooperative disturbance estimation strategy and cooperative control law in practical implementation, real-world flight tests are performed in an indoor flight test environment. The test facility consists of Optitrack motion capture system, ground station, and quadrotor platforms, which have been used to support various research projects (see [31] and [45] for details). A video demonstrating these scenarios is available at https://youtu.be/KIUFwnfmHZk. System and control parameters in the experiments are same as those in the simulation except for the initial condition.

In the experiments, we set the tripod as a landmark fixed on the ground as shown in Fig. 6. To ensure fairness in the comparison experiments with other algorithms, the pipe is pushed until it reaches the landmark and then released so that the initial conditions for the experiments of different algorithms are guaranteed to be the same. In addition, control

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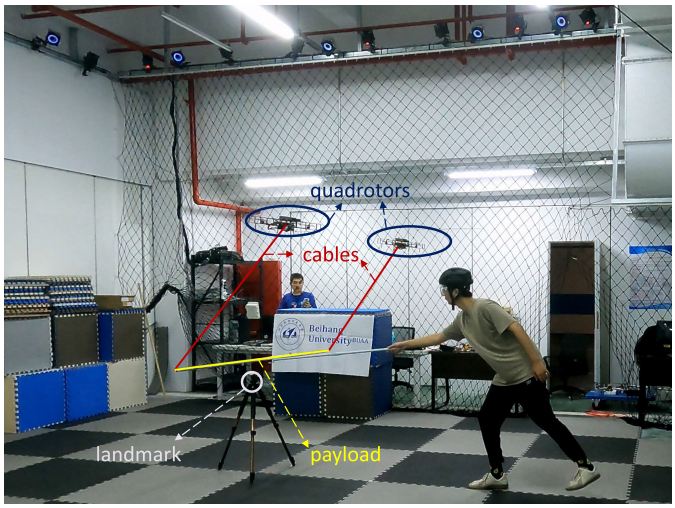

Fig. 6. Experimental schematic.

TABLE III

MEAN AND STD OF ABSOLUTE ERROR AND RSME OF ESTIMATION RESULTS

	Statistics		
	MEAN	STD	RMSE
$p_{01,y}$ (m)	0.070	0.111	0.131
$p_{02,y}$ (m)	0.058	0.110	0.125

parameters for all the algorithms have been carefully selected, and the push-and-release process is repeated and averaged for nine times for each algorithm to have a consistent result. The related experimental data is recorded through the ground station.

Five other methods (NHC in [21], IDA-PBC in [22], MOBADC in [31], PO in [23], FT in [24]) are also chosen for experimental comparison. The experimental results further validate the merits of the controller (41) in payload swing suppression.

A. Attachment Point Estimation

Since the position of the payload $\boldsymbol{p}_0$ and the Euler angles θ_0 and ψ_0 of the payload are measured by the motion capture system, the ground truth of the Y positions $p_{0i,y}$ $(i = 1, 2)$ can be computed through (58). The comparison of the ground truth $p_{0i,y}$ $(i = 1, 2)$ and the estimates $\hat{p}_{0i,y}$ $(i = 1, 2)$ in the experiment is depicted in Fig. 7, where the three peaks of the curves correspond to the moment when the pipe is pushed to the fixed position. The estimation results are further evaluated using absolute errors and Root Mean Square Errors (RMSE). The mean and standard deviation of the absolute errors and the RMSE of the Y positions of the attachment points comparing with the ground truth are shown in Table III. One can conclude from Table III that the proposed swing observer can estimate the positions of the attachment points on the payload with small errors comparing with the ground truth from the motion capture system. This illustrates that our swing observer (36) can provide accurate information about the position of the payload in real time for closed-loop control.

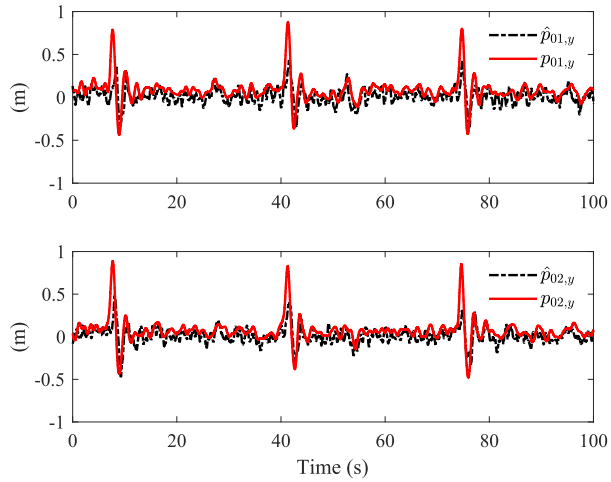

Fig. 7. Estimation results of the Y positions of the attachment points in the experiment.

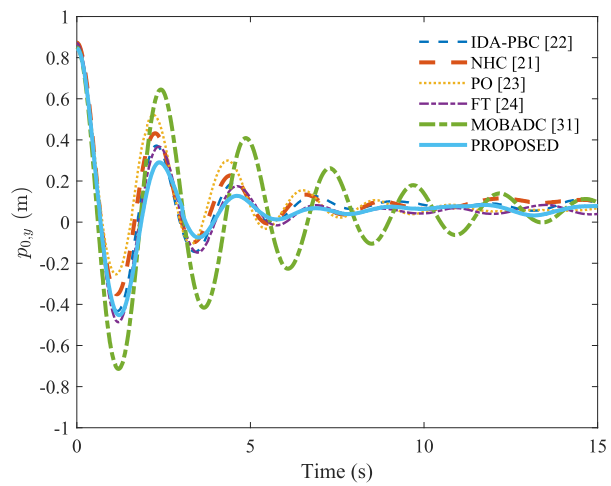

Fig. 8. Time evolution of the CoM position of the payload in the Y direction in the experiment.

TABLE IV

COMPARISON OF THE PERFORMANCE OF PAYLOAD SWING SUPPRESSION

	y_1 (m)	y_2 (m)	y_3 (m)	t_s (s)	χ
IDA-PBC [22]	0.873	0.372	0.190	4.825	1.154
NHC [21]	0.876	0.431	0.227	4.830	1.174
PO [23]	0.872	0.521	0.329	6.640	1.398
FT [24]	0.870	0.367	0.175	4.860	1.206
MOBADC [31]	0.871	0.645	0.409	9.955	2.628
PROPOSED	**0.871**	**0.291**	**0.128**	**2.860**	**0.909**

B. Swing Suppression

In each experiment, we record the time when the payload is released as $t = 0$ and collect the Y position data of the payload in the following thirty seconds. As shown in Fig. 8, the lines show the average Y position of the payload in nine repeated experiment. The performance of the methods in [21], [22], [23], [24], and [31] is also shown in Fig. 8 for comparison. We define y_i as the amplitude of the ith local maximum of the lines in Fig. 8. Here the first three local maxima of the lines from $t = 0$, i.e., y_1, y_2, y_3, are selected for comparison, which are recorded in Table IV. It can be observed from

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the recorded values of y_1 that the payload is released from almost the same Y position, because the differences of these methods in y_1 are millimeter level, which is close to the limits of measurement accuracy of the motion capture system. Based on this observation and the recorded values of y_2 and y_3 in Table IV, it can be inferred that the payload swing magnitude of the FT method decays fastest in the comparison methods [21], [22], [23], [24], [31]. In contrast with the FT method, the proposed method reduces the magnitude y_2 and y_3 by 20.7% and 26.9%, respectively. On top of this, the improvements are even significant at 54.9% and 68.7%, when compared with the MOBADC method. Therefore, it is not difficult to draw the conclusion that the proposed method achieves the fastest convergence of the payload position in the Y direction, which is consistent with the simulation result in Fig. 5.

In addition, two extra indexes are defined to evaluate the performance of the payload swing suppression. One index is the settling time t_s for the magnitude of $p_{0,y}$ to enter the allowable level of residual vibration, which is defined as the section $[-0.15\text{m}, 0.15\text{m}]$, and the other index is the integration of the magnitude of the residual vibration during the settling time period $[0, t_s]$:

$$\chi = \int_0^{t_s} |p_{0,y}(t)|dt \tag{59}$$

The larger the swing magnitude during the settling time period, the larger the index χ will be. Compared with the settling time t_s, the index χ highlights the swing range and the swing frequency of the payload. The comparisons of different methods based on the two indexes (t_s and χ) are also presented in Table IV. As shown in the table, the settling time t_s of the proposed method is less than that of the IDA-PBC method by 40.7%, and even much less than that of the rest of the compared methods. On top of this, as long as the index χ is concerned, the proposed method also achieves the smallest number, which is smaller than the second smallest value (IDA-PBC) by 21.2%. The quantitative results further illustrate the superiority of the controller (41) in rapid oscillation suppression of the payload. In general, we attribute the excellent performance of the proposed method to the refined cooperative disturbance estimation strategy (36) (37) and the cooperative control law (41), which derive the precise estimates of the cable states and utilize them for feedback control, respectively.

VII. Conclusion

In this article, a systematic and complete treatment from system modeling, control design, to experimental verification has been developed for oscillation suppression-enhanced cooperative control of the aerial co-transportation systems. The effectiveness of the proposed refined cooperative disturbance estimation strategy has been verified in the experiments. In addition, it has been demonstrated experimentally that the proposed cooperative controller can achieve better performance than the existing methods thanks to the well-designed refined cooperative disturbance estimation strategy and the cooperative control law. Although the proposed method can be employed to estimate the cable directions even under the effect of model uncertainties and nonlinearity, it requires accurate knowledge of specific model parameters, such as the lengths of the two cables and the mass of the payload. To adapt to the case with uncertain model parameters, the adaptive estimation technique [46] may be employed in the future.

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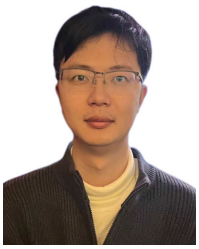

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